Classical entanglement and entropy of two coupled quantum rotors

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Abstract

Duan, Kovner and Skokov showed, for two weakly coupled harmonic oscillators, that the entanglement entropy of a highly excited eigenstate coincides with the Boltzmann entropy of the reduced classical phase-space distribution, provided the elementary phase-space cell is chosen as $h/2$. We carry out the same programme for two rotors on a circle coupled by $\lambda\cos(\theta_1-\theta_2)$, a system that is not Gaussian, whose configuration space is compact, and whose relative motion has a separatrix. All calculations are given in full. Perturbatively we obtain the eigenstate, the energy shift $E^{(2)} = \lambda^2 I/[2\hbar^2(n^2-1)]$ and the density matrix to second order, and show that an $O(\lambda^2)$ normalisation term, invisible in the state at the order to which it is kept, supplies the whole linear entropy and the whole non-logarithmic part of the entanglement entropy. Non-perturbatively we show that every sector of fixed total angular momentum M is exactly a Mathieu problem, periodic or antiperiodic according to the parity of M, with $q = 2\lambda I/\hbar^2$, and that the entanglement entropy is the Shannon entropy of the Fourier coefficients of the corresponding Mathieu function. The reduced classical distribution is obtained in closed form, with an elliptic-integral normalisation; because the angular momentum is quantised, the phase-space cell is fixed at h and the comparison contains no adjustable constant. Deep in the rotation regime the problem reduces to a Wannier–Stark ladder, the Schmidt weights become $J_j^2(\gamma)$ with $\gamma = \lambda I/\hbar^2 n$ up to corrections of order $\lambda/E_{\rm rel}$, and the classical weights become the arcsine law. A momentum-space WKB analysis shows that the quantum entropy lies below the classical one by the universal constant $1-\ln 2$ in the semiclassical limit. In deep rotation we confirm the approach, as $\gamma^{-1/3}$, up to $\gamma = 10^6$; at generic energies the deficit grows towards the same value. The leading logarithms therefore agree, as in the oscillator case, but the exact equality does not survive: the conserved angular momentum fixes the Schmidt basis, so the interference between the two classical branches appears in the entanglement spectrum itself. Both entropies are maximal at the separatrix, where the support of the momentum distribution has a cusp. In deep libration $S_0 = \frac{1}{4}\ln(\lambda I/\hbar^2) + \frac{1}{2}\ln(\pi e/\sqrt{2}) - [8\sqrt{2\lambda I/\hbar^2}\,]^{-1} + O(\hbar^2/\lambda I)$.

Keywords: entanglement entropy, quantum rotors, quantum pendulum, Mathieu equation, classical–quantum correspondence, separatrix, WKB

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1. Introduction

Quantum entanglement has become a working tool well outside quantum information theory [1, 2, 3]. One context in which it has attracted recent attention is high-energy scattering, where it has been argued that the entanglement between the degrees of freedom probed in deeply inelastic scattering and the remainder of the hadron wave function is converted, in the collision, into the Boltzmann entropy of the produced particles [4, 5, 6]. The regime of interest — close to saturation — is precisely the regime in which the classical Colour Glass Condensate description is expected to hold. The juxtaposition raises a sharp question: how much of what is called quantum entanglement in such states is reproduced by a purely classical calculation?

Duan, Kovner and Skokov (DKS) [7] addressed this in the simplest setting, two coupled harmonic oscillators

$$H_c = \tfrac{1}{2}\left(p_x^2 + p_y^2 + \omega^2 x^2 + \Omega^2 y^2 + 2Cxy\right), \qquad C \ll \omega^2 \ll \Omega^2. \tag{1}$$

Classically they integrated the microcanonical distribution over the second oscillator and computed the Boltzmann entropy of the reduced distribution $W(x, p_x)$; quantum mechanically they computed the von Neumann entropy of the reduced density matrix of a highly excited eigenstate in the WKB approximation. The two agree,

$$S_{\mathrm{CE}} = S_E = \ln\left[\frac{\pi C}{\Omega}\frac{\sqrt{E_1 E_2}}{\hbar\omega\Omega}\right], \tag{2}$$

and the reduced density matrix is close to maximally mixed on its support. The conclusion drawn in [7] is that, in the regime relevant to high-energy scattering, such entanglement is of classical origin.

The oscillator pair is special in three respects, any of which might be responsible for the agreement. *(i) It is Gaussian.* Equation (1) is diagonalised by a normal-mode rotation, so the entanglement is entirely a property of that rotation. *(ii) Its phase space is non-compact.* The probability density W is dimensionful, and defining an entropy requires dividing by an elementary phase-space volume Δ that classical mechanics does not supply. The exact equality (2) holds for $\Delta = h/2$, a value fixed by comparison with the quantum answer; any other choice shifts the classical entropy by a constant [7]. *(iii) It has no separatrix.* The orbit topology never changes, so the oscillator pair cannot test the correspondence at a point where one would expect a classical description to be most delicate.

We repeat the analysis for two rotors on a circle,

$$H = \frac{p_1^2}{2I} + \frac{p_2^2}{2I} + \lambda\cos\left(\theta_1 - \theta_2\right), \tag{3}$$

which removes all three restrictions simultaneously. The configuration space is the torus $S^1 \times S^1$, so the momenta are quantised in units of $\hbar$ and the cell is fixed at h; the coupling is a cosine, not bilinear; and the relative motion is a pendulum with a separatrix. At the same time the model is solvable: every sector of fixed total angular momentum is exactly a Mathieu problem.

The main results are the following.

1. The perturbative eigenstate, energy and density matrix to second order (Secs. 3–5), including an $O(\lambda^2)$ normalisation term that is invisible in the state but supplies the whole linear entropy.

2. An exact mapping of every sector onto the Mathieu equation, with the entanglement entropy equal to the Shannon entropy of the Fourier coefficients of a Mathieu function (Sec. 6).
3. The reduced classical distribution in closed form, with no free cell (Sec. 7).
4. An exactly solvable deep-rotation limit in terms of Bessel functions, and a WKB derivation, checked numerically, that the quantum entropy falls below the classical one by $1-\ln 2$ in the semiclassical limit (Secs. 8, 9).
5. The behaviour at the separatrix and in deep libration (Secs. 9, 10).

We have aimed to give every calculation in full, at the level of a graduate course in quantum mechanics; standard integrals are collected in Appendix A and every analytic statement has been checked numerically (Appendix B).

2. The model

2.1. Configuration space and basis

Each rotor is a rigid body free to rotate about a fixed axis, described by an angle $\theta_i \in [0, 2\pi)$ with θ_i and $\theta_i + 2\pi$ identified. The canonical momentum is the angular momentum $p_i = -i\hbar\,\partial/\partial\theta_i$. The Hilbert space is $\mathcal{H} = L^2(S^1) \otimes L^2(S^1)$, and wave functions must be 2π-periodic in each angle.

The eigenfunctions of p_i are $e^{im\theta}$; periodicity, $e^{im(\theta+2\pi)} = e^{im\theta}$, requires $e^{2\pi i m} = 1$, i.e. $m \in \mathbb{Z}$. This is the origin of the quantisation $p_i = \hbar m_i$, which will matter repeatedly. The uncoupled basis is

$$\langle \theta_1, \theta_2 | m_1, m_2 \rangle = \frac{1}{2\pi}\, e^{im_1\theta_1} e^{im_2\theta_2}, \qquad m_1, m_2 \in \mathbb{Z}. \tag{4}$$

Orthonormality follows from

$$\int_0^{2\pi} \mathrm{d}\theta\, e^{i(m-m')\theta} = \begin{cases} 2\pi, & m = m', \\ \dfrac{e^{2\pi i(m-m')} - 1}{i(m-m')} = 0, & m \neq m', \end{cases} \tag{5}$$

so that

$$\langle m_1', m_2' | m_1, m_2 \rangle = \frac{1}{(2\pi)^2} \int_0^{2\pi} \mathrm{d}\theta_1\, e^{i(m_1-m_1')\theta_1} \int_0^{2\pi} \mathrm{d}\theta_2\, e^{i(m_2-m_2')\theta_2} = \delta_{m_1 m_1'}\, \delta_{m_2 m_2'}. \tag{6}$$

Completeness, $\sum_m |m\rangle\langle m| = \mathbb{1}$, is the statement that the Fourier series represents every square-integrable periodic function.

2.2. The free Hamiltonian

In the angle representation

$$H_0 = \frac{p_1^2 + p_2^2}{2I} = -\frac{\hbar^2}{2I}\left(\frac{\partial^2}{\partial\theta_1^2} + \frac{\partial^2}{\partial\theta_2^2}\right). \tag{7}$$

Acting on (4), each derivative brings down im_i, so $\partial_{\theta_i}^2 e^{im_i\theta_i} = -m_i^2 e^{im_i\theta_i}$ and

$$H_0\,|m_1, m_2\rangle = E^{(0)}_{m_1,m_2}\,|m_1, m_2\rangle\,, \qquad E^{(0)}_{m_1,m_2} = \frac{\hbar^2}{2I}\left(m_1^2 + m_2^2\right). \tag{8}$$

2.3. Matrix elements of the coupling

Write $\cos(\theta_1 - \theta_2) = \frac{1}{2}\left[e^{i\theta_1}e^{-i\theta_2} + e^{-i\theta_1}e^{i\theta_2}\right]$. The multiplication operator $e^{i\theta}$ acts on the single-rotor basis as a ladder operator:

$$\left\langle\theta\middle|e^{i\theta}\,m\right\rangle = e^{i\theta}\frac{e^{im\theta}}{\sqrt{2\pi}} = \frac{e^{i(m+1)\theta}}{\sqrt{2\pi}} = \langle\theta|m+1\rangle \quad\Longrightarrow\quad e^{\pm i\theta}\,|m\rangle = |m\pm 1\rangle\,. \tag{9}$$

Hence

$$e^{i(\theta_1-\theta_2)}\,|m_1,m_2\rangle = |m_1+1,m_2-1\rangle\,, \tag{10}$$

$$e^{-i(\theta_1-\theta_2)}\,|m_1,m_2\rangle = |m_1-1,m_2+1\rangle\,, \tag{11}$$

and, using (6),

$$\langle m_1',m_2'|\cos(\theta_1-\theta_2)\,|m_1,m_2\rangle = \frac{1}{2}\left[\delta_{m_1',m_1+1}\delta_{m_2',m_2-1} + \delta_{m_1',m_1-1}\delta_{m_2',m_2+1}\right]. \tag{12}$$

The coupling moves one unit of angular momentum from one rotor to the other. Its matrix is real and symmetric, as it must be for a Hermitian operator in a real basis.

2.4. Conservation of total angular momentum

Let $J = p_1 + p_2 = -i\hbar(\partial_{\theta_1} + \partial_{\theta_2})$. Clearly $[H_0, J] = 0$, since all derivatives commute. For the coupling, act on an arbitrary ψ:

$$[J,\cos(\theta_1-\theta_2)]\,\psi = -i\hbar\,(\partial_{\theta_1}+\partial_{\theta_2})\,[\cos(\theta_1-\theta_2)]\,\psi = -i\hbar\,[-\sin(\theta_1-\theta_2)+\sin(\theta_1-\theta_2)]\,\psi = 0, \tag{13}$$

where the product rule leaves only the derivative of the cosine and the two terms cancel because the cosine depends on $\theta_1 - \theta_2$ alone. Thus $[H, J] = 0$. The eigenvalue of $J/\hbar$,

$$M = m_1 + m_2, \tag{14}$$

is conserved, as is evident from (12), and $\mathcal{H}$ decomposes into invariant sectors $\mathcal{H}_M$.

2.5. Relative quantum number and the sector Hamiltonian

Within $\mathcal{H}_M$ the remaining label is the relative quantum number

$$n = m_1 - m_2, \qquad m_1 = \frac{M+n}{2}, \quad m_2 = \frac{M-n}{2}. \tag{15}$$

Since m_1 and m_2 are integers, $M + n = 2m_1$ is even: $n \equiv M \pmod 2$. Using

$$m_1^2 + m_2^2 = \frac{(M+n)^2 + (M-n)^2}{4} = \frac{2M^2+2n^2}{4} = \frac{M^2+n^2}{2}, \tag{16}$$

the unperturbed energy becomes

$$E^{(0)} = \frac{\hbar^2}{4I}\left(M^2+n^2\right). \tag{17}$$

The first term is the kinetic energy of the centre of mass; the second, of the relative motion. In terms of n, the coupling (12) connects n with $n' = (m_1 \pm 1) - (m_2 \mp 1) = n \pm 2$.

2.6. Exchange symmetry and its consequences

Define the exchange operator $P\,|m_1, m_2\rangle = |m_2, m_1\rangle$, i.e. $P : \theta_1 \leftrightarrow \theta_2$, under which $M \to M$ and $n \to -n$. H_0 is symmetric in the two rotors, and $\cos(\theta_1 - \theta_2) = \cos(\theta_2 - \theta_1)$, so $[H, P] = 0$. Three consequences follow.

(a) Degeneracy. By (17) the energy depends on n only through n^2, so $|m_1, m_2\rangle$ and $|m_2, m_1\rangle$ are degenerate whenever $m_1 \neq m_2$. Only $n = 0$ is non-degenerate.

(b) The order at which the degeneracy is lifted. Each application of V changes n by ± 2, so the shortest chain of matrix elements connecting n to $-n$ has $|n|$ steps. The two members of the degenerate pair are therefore coupled first at order $\lambda^{|n|}$, and the splitting of the doublet is $\delta E \propto \lambda^{|n|}$. Two cases deserve mention. For $|n| = 1$ (which requires M odd) the pair $n = \pm 1$ is coupled directly, at first order; this is the resonance at which non-degenerate perturbation theory fails in Sec. 3. For $|n| = 2$ it fails at second order.

(c) Exact eigenstates are parity eigenstates. For any $\lambda > 0$ the exact eigenstates in a degenerate pair are $(|\Psi_n\rangle \pm |\Psi_{-n}\rangle)/\sqrt{2}$, with $|\Psi_{-n}\rangle = P\,|\Psi_n\rangle$. Non-degenerate perturbation theory about $|m_1, m_2\rangle$ describes one *branch*, $|\Psi_n\rangle$. On times short compared with $\hbar/\delta E$ the branch state is effectively stationary; for $|n| \gtrsim 10$ and small λ, $\hbar/\delta E$ exceeds any physical time scale. We compute both. The relation between their entropies is derived in Sec. 5.7.

2.7. Centre-of-mass and relative variables

For the classical analysis, and to identify the pendulum, introduce

$$\Theta = \frac{\theta_1 + \theta_2}{2}, \qquad \varphi = \theta_1 - \theta_2, \qquad J = p_1 + p_2, \qquad p = \frac{p_1 - p_2}{2}. \tag{18}$$

This transformation is canonical: from $\theta_1 = \Theta + \varphi/2$ and $\theta_2 = \Theta - \varphi/2$,

$$p_1\dot{\theta}_1 + p_2\dot{\theta}_2 = (p_1 + p_2)\dot{\Theta} + \frac{p_1 - p_2}{2}\dot{\varphi} = J\dot{\Theta} + p\dot{\varphi}, \tag{19}$$

so (Θ, J) and (φ, p) are conjugate pairs. Inverting, $p_1 = J/2 + p$ and $p_2 = J/2 - p$, whence $p_1^2 + p_2^2 = J^2/2 + 2p^2$ and

$$H = \frac{J^2}{4I} + \underbrace{\frac{p^2}{I} + \lambda\cos\varphi}_{H_{\rm rel}}. \tag{20}$$

The relative motion is a pendulum with moment of inertia $\mu = I/2$ (the reduced moment of inertia of two equal rotors), since $p^2/I = p^2/2\mu$. The potential $\lambda\cos\varphi$ has its minimum $-\lambda$ at $\varphi = \pi$ and its maximum $+\lambda$ at $\varphi = 0$, so the separatrix is at

$$E_{\rm rel} \equiv E - \frac{J^2}{4I} = \lambda. \tag{21}$$

For $|E_{\rm rel}| < \lambda$ the relative angle librates about $\varphi = \pi$ (the rotors are phase-locked, anti-aligned); for $E_{\rm rel} > \lambda$ it circulates. Quantum mechanically, $J = \hbar M$, $p = \hbar n/2$, and (20) reproduces (17) at $\lambda = 0$.

3. Perturbation theory

3.1. Rayleigh–Schrödinger theory in intermediate normalisation

For completeness we derive the formulas we use. Write $H = H_0 + V$ with $V = \lambda\cos(\theta_1 - \theta_2)$, let $|0\rangle$ be a non-degenerate eigenstate of H_0 with energy E_0, and denote the other eigenstates by $|k\rangle$, with $V_{kl} \equiv \langle k|\, V\, |l\rangle$. Expand

$$\left|\widetilde{\Psi}\right\rangle = |0\rangle + \left|\psi^{(1)}\right\rangle + \left|\psi^{(2)}\right\rangle + \ldots, \qquad E = E_0 + E^{(1)} + E^{(2)} + \ldots, \tag{22}$$

with $\left|\psi^{(k)}\right\rangle, E^{(k)} = O(\lambda^k)$, and impose *intermediate normalisation*

$$\left\langle 0\middle|\widetilde{\Psi}\right\rangle = 1 \quad \Longleftrightarrow \quad \left\langle 0\middle|\psi^{(k)}\right\rangle = 0 \quad (k \geq 1). \tag{23}$$

Substituting into $(H_0 + V)\left|\widetilde{\Psi}\right\rangle = E\left|\widetilde{\Psi}\right\rangle$ and collecting powers of λ:

$$O(\lambda^1): \quad (H_0 - E_0)\left|\psi^{(1)}\right\rangle = \left(E^{(1)} - V\right)|0\rangle\,, \tag{24}$$

$$O(\lambda^2): \quad (H_0 - E_0)\left|\psi^{(2)}\right\rangle = \left(E^{(1)} - V\right)\left|\psi^{(1)}\right\rangle + E^{(2)}\,|0\rangle\,. \tag{25}$$

Projecting (24) onto $\langle 0|$ kills the left-hand side, because $\langle 0|\,(H_0 - E_0) = 0$; hence $E^{(1)} = V_{00}$. Projecting onto $\langle k|$, $k \neq 0$, gives $(E_k - E_0)\left\langle k\middle|\psi^{(1)}\right\rangle = -V_{k0}$, i.e.

$$\left|\psi^{(1)}\right\rangle = \sum_{k\neq 0} \frac{V_{k0}}{E_0 - E_k}\,|k\rangle\,. \tag{26}$$

Projecting (25) onto $\langle 0|$ and using (23),

$$E^{(2)} = \langle 0|\, V\left|\psi^{(1)}\right\rangle = \sum_{k\neq 0} \frac{|V_{k0}|^2}{E_0 - E_k}, \tag{27}$$

and onto $\langle k|$, $k \neq 0$, after inserting $\sum_l |l\rangle\langle l|$ in $V\left|\psi^{(1)}\right\rangle$ and using $\left\langle 0\middle|\psi^{(1)}\right\rangle = 0$,

$$\left|\psi^{(2)}\right\rangle = \sum_{k\neq 0}\left[\sum_{l\neq 0} \frac{V_{kl}V_{l0}}{(E_0 - E_k)(E_0 - E_l)} - \frac{V_{00}V_{k0}}{(E_0 - E_k)^2}\right]|k\rangle\,. \tag{28}$$

The normalised state is $|\Psi\rangle = \mathcal{N}\left|\widetilde{\Psi}\right\rangle$ with $\mathcal{N} = \left\langle\widetilde{\Psi}\middle|\widetilde{\Psi}\right\rangle^{-1/2}$.

3.2. Unperturbed level and its neighbours

We expand about $|0\rangle \equiv |m_1, m_2\rangle$ in sector M, with relative quantum number n. By (12) the only non-zero matrix elements of V from $|0\rangle$ are

$$V_{\pm,0} = \frac{\lambda}{2}, \qquad |\pm\rangle \equiv |m_1 \pm 1, m_2 \mp 1\rangle \quad (n \to n \pm 2). \tag{29}$$

In particular $V_{00} = 0$. The energy differences follow from (8):

$$\begin{aligned} E_0 - E_\pm &= \frac{\hbar^2}{2I}\left[m_1^2 + m_2^2 - (m_1 \pm 1)^2 - (m_2 \mp 1)^2\right] \\ &= \frac{\hbar^2}{2I}\left[m_1^2 + m_2^2 - \left(m_1^2 \pm 2m_1 + 1\right) - \left(m_2^2 \mp 2m_2 + 1\right)\right] \\ &= \frac{\hbar^2}{2I}\left[\mp 2(m_1 - m_2) - 2\right] = \frac{\hbar^2}{I}\left[\mp n - 1\right], \end{aligned} \tag{30}$$

that is,

$$E_0 - E_+ = -\frac{\hbar^2}{I}(n+1), \qquad E_0 - E_- = +\frac{\hbar^2}{I}(n-1). \tag{31}$$

The same result follows from (17): $\frac{\hbar^2}{4I}[n^2 - (n \pm 2)^2] = \frac{\hbar^2}{4I}(\mp 4n - 4)$. For later use, the next-nearest neighbours $|\pm\pm\rangle \equiv |m_1 \pm 2, m_2 \mp 2\rangle$ $(n \to n \pm 4)$ have

$$E_0 - E_{\pm\pm} = \frac{\hbar^2}{4I}\left[n^2 - (n \pm 4)^2\right] = \frac{\hbar^2}{4I}\left(\mp 8n - 16\right) = \mp\frac{2\hbar^2}{I}(n \pm 2). \tag{32}$$

3.3. First order

Energy. $E^{(1)} = V_{00} = \lambda \langle m_1, m_2 | \cos(\theta_1 - \theta_2) | m_1, m_2 \rangle = 0$, because every term in (12) requires $m_1' \neq m_1$. This holds for every level, exactly.

State. From (26) with (31),

$$\left|\psi^{(1)}\right\rangle = \frac{\lambda/2}{-\hbar^2(n+1)/I}\left|+\right\rangle + \frac{\lambda/2}{\hbar^2(n-1)/I}\left|-\right\rangle. \tag{33}$$

It is convenient to introduce the dimensionless coupling

$$\boxed{\alpha \equiv \frac{\lambda I}{2\hbar^2}} \tag{34}$$

the ratio of the coupling to twice the rotational level spacing scale $\hbar^2/I$. Then

$$\left|\psi^{(1)}\right\rangle = c_+\left|+\right\rangle + c_-\left|-\right\rangle, \qquad c_+ = -\frac{\alpha}{n+1}, \qquad c_- = +\frac{\alpha}{n-1}. \tag{35}$$

This requires $n \neq \pm 1$, the first-order resonance of Sec. 2.6(b).

3.4. Second-order energy

From (27) with the two non-zero V_{k0},

$$\begin{aligned} E^{(2)} &= \left(\frac{\lambda}{2}\right)^2\left[\frac{1}{E_0 - E_+} + \frac{1}{E_0 - E_-}\right] = \frac{\lambda^2 I}{4\hbar^2}\left[-\frac{1}{n+1} + \frac{1}{n-1}\right] \\ &= \frac{\lambda^2 I}{4\hbar^2}\cdot\frac{-(n-1) + (n+1)}{(n+1)(n-1)} = \frac{\lambda^2 I}{4\hbar^2}\cdot\frac{2}{n^2-1}, \end{aligned} \tag{36}$$

so

$$\boxed{E^{(2)} = \frac{\lambda^2 I}{2\hbar^2\,(n^2-1)} = \frac{2\hbar^2}{I}\,\frac{\alpha^2}{n^2-1}} \tag{37}$$

For $n = 0$ this is $-\lambda^2 I/2\hbar^2 < 0$: the lowest state of the sector is pushed down by both neighbours, which lie above it. For $|n| \geq 2$ it is positive: the neighbour $n \to n-2$ (towards equal angular momenta) lies below and dominates. Equation (37) agrees with exact diagonalisation to six significant figures at $\lambda = 0.02$ (Appendix B).

3.5. Second-order state

Since $V_{00} = 0$ the second term of (28) vanishes. In the first term, $V_{kl}V_{l0} \neq 0$ requires $l = \pm$ (one step from n) and then k one step from l, i.e. $k \in \{n, n \pm 4\}$. The value $k = n$ is the state $|0\rangle$ itself, which is excluded from the sum whichever intermediate state $l = \pm$ leads back to it. Only $k = \pm\pm$ remain, each reached through a single path:

$$\langle ++|\psi^{(2)}\rangle = \frac{V_{++,+}\,V_{+,0}}{(E_0 - E_{++})(E_0 - E_+)} = \frac{(\lambda/2)^2}{\left[-\frac{2\hbar^2}{I}(n+2)\right]\left[-\frac{\hbar^2}{I}(n+1)\right]} = \frac{\lambda^2 I^2}{8\hbar^4(n+1)(n+2)}, \tag{38}$$

$$\langle --|\psi^{(2)}\rangle = \frac{(\lambda/2)^2}{\left[\frac{2\hbar^2}{I}(n-2)\right]\left[\frac{\hbar^2}{I}(n-1)\right]} = \frac{\lambda^2 I^2}{8\hbar^4(n-1)(n-2)}. \tag{39}$$

In terms of α, using $\lambda^2 I^2/\hbar^4 = 4\alpha^2$,

$$|\psi^{(2)}\rangle = c_{++}\,|++\rangle + c_{--}\,|--\rangle\,, \qquad c_{++} = \frac{\alpha^2}{2(n+1)(n+2)}, \qquad c_{--} = \frac{\alpha^2}{2(n-1)(n-2)}. \tag{40}$$

This additionally requires $n \neq \pm 2$, the second-order resonance.

3.6. Normalisation

Because $\langle 0|\psi^{(1)}\rangle = \langle 0|\psi^{(2)}\rangle = 0$ and $|\pm\rangle$, $|\pm\pm\rangle$ are orthonormal,

$$\left\langle \widetilde{\Psi}\middle|\widetilde{\Psi}\right\rangle = 1 + \underbrace{c_+^2 + c_-^2}_{O(\lambda^2)} + \underbrace{c_{++}^2 + c_{--}^2}_{O(\lambda^4)} + \ldots \tag{41}$$

There is no $O(\lambda)$ term, precisely because of intermediate normalisation. Defining

$$\boxed{\varepsilon_+ \equiv c_+^2 = \frac{\alpha^2}{(n+1)^2} = \frac{\lambda^2 I^2}{4\hbar^4(n+1)^2}, \qquad \varepsilon_- \equiv c_-^2 = \frac{\alpha^2}{(n-1)^2} = \frac{\lambda^2 I^2}{4\hbar^4(n-1)^2}} \tag{42}$$

and $\epsilon \equiv \varepsilon_+ + \varepsilon_-$, the binomial series $(1+x)^{-1/2} = 1 - \frac{1}{2}x + \frac{3}{8}x^2 - \ldots$ gives

$$\mathcal{N} = (1+\epsilon)^{-1/2} + O(\lambda^4) = 1 - \frac{\epsilon}{2} + O(\lambda^4), \qquad \mathcal{N}^2 = 1 - \epsilon + O(\lambda^4). \tag{43}$$

$\mathcal{N} - 1$ is of second order. It does not create any new basis state; it only rescales the amplitude of $|0\rangle$.

3.7. The eigenstate to second order

Multiplying out $\mathcal{N}\left[|0\rangle + |\psi^{(1)}\rangle + |\psi^{(2)}\rangle\right]$ and discarding terms of order λ^3 (in particular $\mathcal{N}c_\pm = c_\pm + O(\lambda^3)$),

$$\begin{aligned}|\Psi\rangle = {} & \left[1 - \frac{\lambda^2 I^2}{8\hbar^4}\left(\frac{1}{(n+1)^2} + \frac{1}{(n-1)^2}\right)\right]|m_1, m_2\rangle \\ & - \frac{\lambda I}{2\hbar^2(n+1)}\,|m_1+1, m_2-1\rangle + \frac{\lambda I}{2\hbar^2(n-1)}\,|m_1-1, m_2+1\rangle \\ & + \frac{\lambda^2 I^2}{8\hbar^4(n+1)(n+2)}\,|m_1+2, m_2-2\rangle + \frac{\lambda^2 I^2}{8\hbar^4(n-1)(n-2)}\,|m_1-2, m_2+2\rangle + O(\lambda^3).\end{aligned} \tag{44}$$

All four non-trivial amplitudes agree with exact diagonalisation to four significant figures for $\lambda I/\hbar^2 \in [0.02, 0.2]$ at $n = 7$ (Table B.1).

3.8. Domain of validity

Equation (44) requires (i) $n \notin \{\pm 1, \pm 2\}$, the resonances at which the exchange partner is reached at first or second order; and (ii) small admixtures, $|c_\pm| \ll 1$, i.e.

$$\alpha \ll |n \pm 1|. \tag{45}$$

For $|n| \gg 1$ this reads $\lambda \ll 2\hbar^2|n|/I$: the coupling must be small compared with the energy spacing $\hbar^2|n|/I$ between neighbouring levels of the relative motion, not merely compared with $\hbar^2/I$. Highly excited states are therefore more perturbative, not less.

4. The density matrix to second order

4.1. Why the normalisation cannot be dropped

Equation (44) is accurate to $O(\lambda)$ in its non-trivial amplitudes, and to that order the normalisation factor is unity. One is tempted to set $\mathcal{N} = 1$ and proceed; this is correct for the state, and incorrect for the density matrix.

Write the state schematically as $|\Psi\rangle = a_0\,|0\rangle + \lambda\,|\psi^{(1)}\rangle + \lambda^2\,|\psi^{(2)}\rangle$ with $a_0 = \mathcal{N} = 1 - \frac{1}{2}\epsilon$. The density matrix $\rho = |\Psi\rangle\langle\Psi|$ is bilinear in the amplitudes, so its $|0\rangle\langle 0|$ element is

$$a_0^2 = \left(1 - \tfrac{1}{2}\epsilon\right)^2 = 1 - \epsilon + O(\lambda^4). \tag{46}$$

The $O(\lambda^2)$ shift of an $O(\lambda^0)$ amplitude has become an $O(\lambda^2)$ term in ρ — the same order as $\lambda^2\,|\psi^{(1)}\rangle\langle\psi^{(1)}|$, which no one would think of dropping. Note also that the factor $-\frac{1}{2}$ in $\mathcal{N}$ has become -1: the shift multiplies $|0\rangle$ and $\langle 0|$ both. Explicitly,

$$\rho = |0\rangle\langle 0| + \left(|\psi^{(1)}\rangle\langle 0| + \text{h.c.}\right) + \underbrace{|\psi^{(1)}\rangle\langle\psi^{(1)}|}_{O(\lambda^2)} + \underbrace{\left(|\psi^{(2)}\rangle\langle 0| + \text{h.c.}\right)}_{O(\lambda^2)} - \underbrace{\langle\psi^{(1)}|\psi^{(1)}\rangle\,|0\rangle\langle 0|}_{O(\lambda^2),\ \text{from } \mathcal{N}^2} + O(\lambda^3). \tag{47}$$

Taking the trace and using $\langle 0|\psi^{(k)}\rangle = 0$, $\operatorname{Tr}\rho = 1 + \langle\psi^{(1)}|\psi^{(1)}\rangle + 0 - \langle\psi^{(1)}|\psi^{(1)}\rangle = 1$. Without the last term, $\operatorname{Tr}\rho = 1 + \epsilon$: the object would not be a density matrix at the order at which it is being used.

4.2. Products of amplitudes

Label the five amplitudes of (44) as

$$a_0 = 1 - \tfrac{1}{2}\epsilon, \quad a_\pm = c_\pm = O(\lambda), \quad a_{\pm\pm} = c_{\pm\pm} = O(\lambda^2). \tag{48}$$

Since all amplitudes are real, $\rho_{ij} = a_i a_j$. The 25 products sort by order as follows:

product	number of entries	order	kept?
$a_0 a_0 = 1 - \epsilon + O(\lambda^4)$	1	$1 + \lambda^2$	yes
$a_0 a_\pm = c_\pm + O(\lambda^3)$	4	λ	yes
$a_\pm a_\pm = \varepsilon_\pm$, $a_+ a_- = c_+ c_-$	4	λ^2	yes
$a_0 a_{\pm\pm} = c_{\pm\pm} + O(\lambda^4)$	4	λ^2	yes
$a_\pm a_{\pm\pm}$, $a_\pm a_{\mp\mp}$	8	λ^3	no
$a_{\pm\pm} a_{\pm\pm}$, $a_{++} a_{--}$	4	λ^4	no

The count is $1 + 4 + 4 + 4 + 8 + 4 = 25$. Thirteen entries survive.

4.3. The full density matrix

Collecting the thirteen entries and using $c_+c_- = -\alpha^2/[(n+1)(n-1)] = -\lambda^2 I^2/[4\hbar^4(n+1)(n-1)]$,

$$
\begin{aligned}
\rho = {} & |m_1, m_2\rangle\langle m_1, m_2| \\
& - \frac{\lambda I}{2\hbar^2(n+1)}\Big[\,|m_1+1, m_2-1\rangle\langle m_1, m_2| + |m_1, m_2\rangle\langle m_1+1, m_2-1|\,\Big] \\
& + \frac{\lambda I}{2\hbar^2(n-1)}\Big[\,|m_1-1, m_2+1\rangle\langle m_1, m_2| + |m_1, m_2\rangle\langle m_1-1, m_2+1|\,\Big] \\
& + \frac{\lambda^2 I^2}{4\hbar^4(n+1)^2}\,|m_1+1, m_2-1\rangle\langle m_1+1, m_2-1| \\
& + \frac{\lambda^2 I^2}{4\hbar^4(n-1)^2}\,|m_1-1, m_2+1\rangle\langle m_1-1, m_2+1| \\
& - \frac{\lambda^2 I^2}{4\hbar^4(n+1)(n-1)}\Big[\,|m_1-1, m_2+1\rangle\langle m_1+1, m_2-1| + |m_1+1, m_2-1\rangle\langle m_1-1, m_2+1|\,\Big] \\
& + \frac{\lambda^2 I^2}{8\hbar^4(n+1)(n+2)}\Big[\,|m_1+2, m_2-2\rangle\langle m_1, m_2| + |m_1, m_2\rangle\langle m_1+2, m_2-2|\,\Big] \\
& + \frac{\lambda^2 I^2}{8\hbar^4(n-1)(n-2)}\Big[\,|m_1-2, m_2+2\rangle\langle m_1, m_2| + |m_1, m_2\rangle\langle m_1-2, m_2+2|\,\Big] \\
& - \frac{\lambda^2 I^2}{4\hbar^4}\left[\frac{1}{(n+1)^2} + \frac{1}{(n-1)^2}\right]|m_1, m_2\rangle\langle m_1, m_2| \\
& + O(\lambda^3).
\end{aligned}
\tag{49}
$$

Line nine is the normalisation term of (47); its coefficient is $\epsilon = \varepsilon_+ + \varepsilon_-$, twice the coefficient that $|0\rangle$ carries in the state (44). Lines two and three are the first-order coherences, lines four and five the second-order populations, line six the second-order coherence, and lines seven and eight the interference of the second-order state with $|0\rangle$.

4.4. Trace and purity

Only diagonal elements contribute to the trace, and the five basis states in (49) are orthonormal, so

$$\operatorname{Tr}\rho = (1-\epsilon) + \varepsilon_+ + \varepsilon_- = 1. \tag{50}$$

For a pure state $\operatorname{Tr}\rho^2 = \sum_{ij}\rho_{ij}^2 = \left(\sum_i a_i^2\right)^2 = 1$. To second order the only contributions are the $|0\rangle\langle 0|$ element and the four first-order coherences:

$$\operatorname{Tr}\rho^2 = (1-\epsilon)^2 + 2\left(c_+^2 + c_-^2\right) + O(\lambda^4) = 1 - 2\epsilon + 2\epsilon + O(\lambda^4) = 1. \tag{51}$$

Both identities fail if the last line of (49) is omitted, or carried with the amplitude coefficient $\epsilon/2$ instead of ϵ.

5. Reduced density matrix and entanglement at weak coupling

5.1. The partial trace

The reduced density matrix of rotor 1 is $\rho_1 = \operatorname{Tr}_2\rho = \sum_{k\in\mathbb{Z}}\langle k|_2\ \rho\ |k\rangle_2$. On a single dyad,

$$\operatorname{Tr}_2|a,b\rangle\langle c,d| = \sum_k |a\rangle\,\langle c|\ \langle k|b\rangle\,\langle d|k\rangle = |a\rangle\,\langle c|\ \langle d|b\rangle = \delta_{bd}\ |a\rangle\langle c|\,, \tag{52}$$

where the completeness relation $\sum_k |k\rangle\langle k| = \mathbb{1}$ was used in the second step. Only dyads whose rotor-2 labels coincide survive.

5.2. Term-by-term evaluation

Applying (52) to each line of (49):

dyad in (49)	rotor-2 labels	survives?
$\|m_1, m_2\rangle\langle m_1, m_2\|$ (lines 1 and 9)	m_2 and m_2	**yes** $\to \|m_1\rangle\langle m_1\|$
$\|m_1 \pm 1, m_2 \mp 1\rangle\langle m_1, m_2\|$ (lines 2, 3)	$m_2 \mp 1$ and m_2	no
$\|m_1 \pm 1, m_2 \mp 1\rangle\langle m_1 \pm 1, m_2 \mp 1\|$ (lines 4, 5)	equal	**yes** $\to \|m_1 \pm 1\rangle\langle m_1 \pm 1\|$
$\|m_1 + 1, m_2 - 1\rangle\langle m_1 - 1, m_2 + 1\|$ (line 6)	$m_2 - 1$ and $m_2 + 1$	no
$\|m_1 \pm 2, m_2 \mp 2\rangle\langle m_1, m_2\|$ (lines 7, 8)	$m_2 \mp 2$ and m_2	no

All coherences vanish. This is a consequence of angular-momentum conservation: every state admixed by the coupling differs from $|0\rangle$ in m_2 by exactly as much as in m_1, so distinct m_1 values are always paired with distinct, mutually orthogonal m_2 values. The second-order state drops out entirely; the normalisation term survives because it is attached to the one dyad whose rotor-2 labels agree.

5.3. Result

$$\boxed{\rho_1 = (1 - \varepsilon_+ - \varepsilon_-)\,|m_1\rangle\langle m_1| + \varepsilon_+\,|m_1+1\rangle\langle m_1+1| + \varepsilon_-\,|m_1-1\rangle\langle m_1-1| + O(\lambda^4)} \tag{53}$$

The error is $O(\lambda^4)$ rather than $O(\lambda^3)$ because the $O(\lambda^3)$ terms of ρ are all off-diagonal in m_2. The eigenvalues are

$$p_0 = 1 - \varepsilon_+ - \varepsilon_-, \qquad p_+ = \varepsilon_+, \qquad p_- = \varepsilon_-, \tag{54}$$

and the state (44) is, to $O(\lambda)$, already in Schmidt form,

$$|\Psi\rangle = \sqrt{p_0}\;|m_1\rangle\,|m_2\rangle + \mathrm{sgn}(c_+)\sqrt{p_+}\;|m_1+1\rangle\,|m_2-1\rangle + \mathrm{sgn}(c_-)\sqrt{p_-}\;|m_1-1\rangle\,|m_2+1\rangle + \ldots, \tag{55}$$

with Schmidt weights that sum to one only because p_0 carries the normalisation correction. More generally, conservation of J implies that in *any* eigenstate in sector M each m_1 is paired with the unique $m_2 = M - m_1$, so the Schmidt basis is always the angular-momentum basis and ρ_1 is exactly diagonal at every order. We return to this in Sec. 6.

5.4. Entanglement entropy

With $S = -\operatorname{Tr}\rho_1 \ln\rho_1 = -\sum_i p_i \ln p_i$, treat the three terms separately. For the large eigenvalue, use $\ln(1-\epsilon) = -\epsilon - \frac{1}{2}\epsilon^2 - \frac{1}{3}\epsilon^3 - \ldots$:

$$-p_0 \ln p_0 = (1-\epsilon)\left(\epsilon + \tfrac{1}{2}\epsilon^2 + \ldots\right) = \epsilon + \tfrac{1}{2}\epsilon^2 - \epsilon^2 + O(\epsilon^3) = \epsilon - \tfrac{1}{2}\epsilon^2 + O(\epsilon^3). \tag{56}$$

The small eigenvalues contribute $-\varepsilon_\pm \ln\varepsilon_\pm = \varepsilon_\pm \ln(1/\varepsilon_\pm)$, which cannot be expanded further. Hence

$$\boxed{S = \varepsilon_+ \ln\frac{1}{\varepsilon_+} + \varepsilon_- \ln\frac{1}{\varepsilon_-} + (\varepsilon_+ + \varepsilon_-) + O(\epsilon^2)} \tag{57}$$

The non-logarithmic term $\varepsilon_+ + \varepsilon_-$ is the contribution (56) of p_0; it exists only because $p_0 = 1-\epsilon$ rather than 1. Dropping the normalisation term would remove it entirely, an error of relative size $1/[1+\ln(1/\epsilon)]$, which decays only logarithmically as $\lambda \to 0$. At $n = 0$, where $\varepsilon_+ = \varepsilon_- = \alpha^2$, it is 9.7% at $\lambda I/\hbar^2 = 0.02$ and 14% at 0.2 (Fig. 1).

5.5. Highly excited relative motion

For $|n| \gg 1$, expand $(n \pm 1)^{-2} = n^{-2}\left(1 \mp \frac{2}{n} + \frac{3}{n^2} \mp \dots\right)$, so that

$$\varepsilon_+ + \varepsilon_- = \frac{2\alpha^2}{n^2}\left(1 + \frac{3}{n^2} + \dots\right), \qquad \varepsilon_+ \approx \varepsilon_- \approx \frac{\alpha^2}{n^2}. \tag{58}$$

At leading order

$$S \simeq 2 \cdot \frac{\alpha^2}{n^2} \ln \frac{n^2}{\alpha^2} + \frac{2\alpha^2}{n^2} = \frac{4\alpha^2}{n^2}\left[\ln \frac{n}{\alpha} + \frac{1}{2}\right]. \tag{59}$$

Restoring units with $4\alpha^2 = \lambda^2 I^2/\hbar^4$ and $n/\alpha = 2\hbar^2 n/\lambda I$,

$$\boxed{S \simeq \left(\frac{\lambda I}{\hbar^2 n}\right)^2 \left[\ln \frac{2\hbar^2 n}{\lambda I} + \frac{1}{2}\right]\left[1 + O(n^{-2})\right]} \tag{60}$$

The factor 2 inside the logarithm is easily lost and changes the result by about 10% at $n = 12$–30; with it, (60) agrees with the full expression (57) to 2% at $n = 12$ and 0.3% at $n = 30$, as the $3/n^2$ correction predicts. The entropy falls as $n^{-2}\ln n$: a highly excited relative rotation is weakly entangled because the level spacing $\hbar^2 n/I$ grows while the coupling does not.

5.6. Linear and Rényi entropies

The linear entropy requires no logarithms:

$$S_L = 1 - \operatorname{Tr}\rho_1^2 = 1 - \left[(1-\epsilon)^2 + \varepsilon_+^2 + \varepsilon_-^2\right] = 2\epsilon - \epsilon^2 - \varepsilon_+^2 - \varepsilon_-^2 = 2\left(\varepsilon_+ + \varepsilon_-\right) + O(\epsilon^2). \tag{61}$$

At leading order it is generated *entirely* by the normalisation term: with $p_0 = 1$ one would find $\operatorname{Tr}\rho_1^2 = 1 + \varepsilon_+^2 + \varepsilon_-^2 > 1$ and $S_L < 0$. For the Rényi entropies $S_q = (1-q)^{-1}\ln\sum_i p_i^q$,

$$\sum_i p_i^q = (1-\epsilon)^q + \varepsilon_+^q + \varepsilon_-^q = 1 - q\epsilon + \varepsilon_+^q + \varepsilon_-^q + O(\epsilon^2). \tag{62}$$

For $q > 1$ the terms $\varepsilon_\pm^q$ are of higher order than ϵ, and $S_q \simeq q\epsilon/(q-1)$, again entirely from the normalisation term; $q = 2$ reproduces (61). For $q < 1$ the terms $\varepsilon_\pm^q$ dominate and $S_q \simeq (\varepsilon_+^q + \varepsilon_-^q)/(1-q)$.

5.7. Exact parity eigenstates

For $m_1 \neq m_2$ the exact eigenstate is $|\Psi_\pm\rangle = (|\Psi_n\rangle \pm |\Psi_{-n}\rangle)/\sqrt{2}$ (Sec. 2.6). Then

$$\rho_1 = \tfrac{1}{2}\operatorname{Tr}_2|\Psi_n\rangle\langle\Psi_n| + \tfrac{1}{2}\operatorname{Tr}_2|\Psi_{-n}\rangle\langle\Psi_{-n}| \pm \tfrac{1}{2}\left[\operatorname{Tr}_2|\Psi_n\rangle\langle\Psi_{-n}| + \text{h.c.}\right]. \tag{63}$$

By (52) a cross term needs a component of $|\Psi_n\rangle$ and a component of $|\Psi_{-n}\rangle$ with the same rotor-2 label. The branch $|\Psi_n\rangle$ has rotor-2 labels $m_2 + s$ with amplitudes $O(\lambda^{|s|})$; $|\Psi_{-n}\rangle = P|\Psi_n\rangle$ has rotor-2 labels $m_1 + t$ with amplitudes $O(\lambda^{|t|})$. Equal labels require $s - t = n$, so every cross

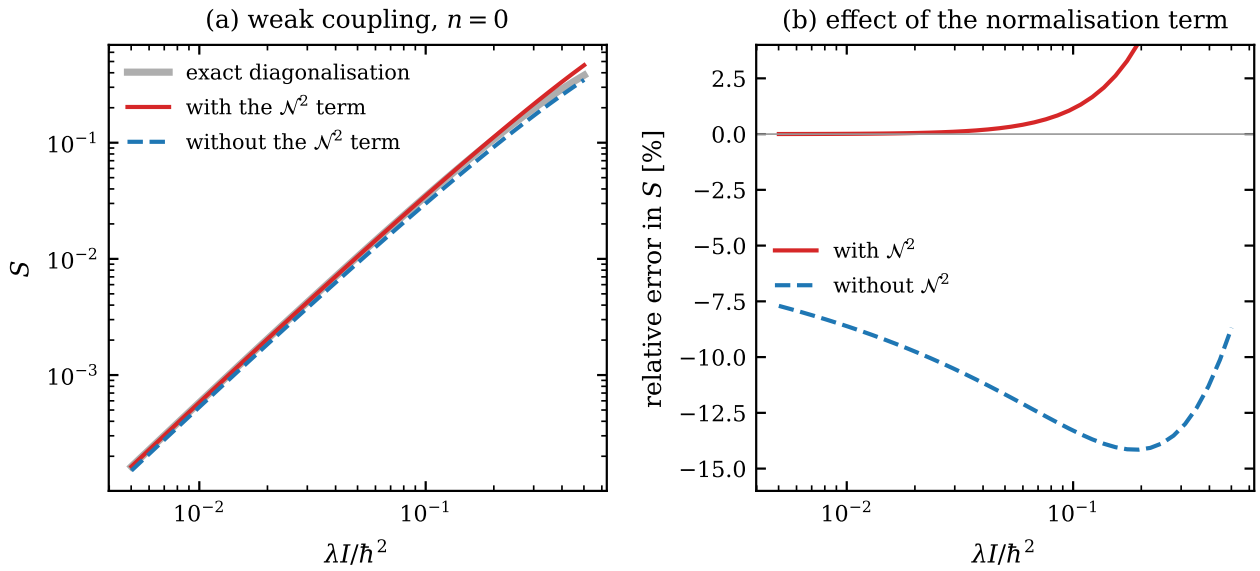


Figure 1: (a) Entanglement entropy at weak coupling for $n = 0$ ($m_1 = m_2 = 4$): exact diagonalisation (grey), Eq. (57) (red), and the same expression with the normalisation term of (49) removed (blue dashed). (b) Relative error of the two truncations. With the term the error is 0.05% at $\lambda I/\hbar^2 = 0.02$ and 1% at 0.1, decreasing as λ^2; without it the deficit is 10–14% and does not vanish as $\lambda \to 0$.

term is of order $\lambda^{|s|+|t|} \geq \lambda^{|n|}$, beyond the order kept when $|n| \geq 3$. The two remaining terms are supported on rotor-1 labels $\{m_1, m_1 \pm 1\}$ and $\{m_2, m_2 \pm 1\}$ respectively, which are disjoint for $|n| \geq 3$. The spectrum of ρ_1 is therefore $\{\frac{1}{2}p_0, \frac{1}{2}p_+, \frac{1}{2}p_-\}$ taken twice, and

$$S_{\text{exact}} = -2\sum_i \tfrac{1}{2}p_i \ln\left(\tfrac{1}{2}p_i\right) = \ln 2 - \sum_i p_i \ln p_i = \ln 2 + S_{\text{branch}}. \tag{64}$$

The extra $\ln 2$ is the entropy of the exchange superposition; it is not generated by the dynamics of the coupling and vanishes on time scales short compared with $\hbar/\delta E \propto \lambda^{-|n|}$.

5.8. Comparison with the oscillator ground state

The structure of (57) is that of the ground state of the oscillator pair (1), which [7] quote as $S_E^0 = -(1-f)\ln(1-f) - f\ln f$ with $f = C^2/4\omega\Omega^3$. It is instructive to rederive f by the method used here. With $x = \sqrt{\hbar/2\omega}\,(a + a^\dagger)$ and $y = \sqrt{\hbar/2\Omega}\,(b + b^\dagger)$, the coupling Cxy connects $|0,0\rangle$ only to $|1,1\rangle$, with $\langle 1,1|\,Cxy\,|0,0\rangle = C\hbar/(2\sqrt{\omega\Omega})$ and energy denominator $-\hbar(\omega+\Omega)$. Thus

$$c_{11} = -\frac{C}{2\sqrt{\omega\Omega}\,(\omega+\Omega)}, \qquad f = c_{11}^2 = \frac{C^2}{4\omega\Omega(\omega+\Omega)^2} \xrightarrow{\Omega \gg \omega} \frac{C^2}{4\omega\Omega^3}, \tag{65}$$

in agreement with [7] (and with the exact Gaussian result to 0.1% at $\omega = 1$, $\Omega = 10$, $C = 0.5$). The oscillator has one admixture channel and a single depleted eigenvalue $1 - f$; the rotors have two channels, $n \to n \pm 2$, with unequal weights. In both cases the term $-(1-f)\ln(1-f)$ is the normalisation contribution.

6. Beyond perturbation theory: the Mathieu equation

6.1. The sector Hamiltonian

Inside sector M label the states by $m \equiv m_1$, so $m_2 = M - m$. Completing the square,

$$m^2 + (M-m)^2 = 2m^2 - 2Mm + M^2 = 2\left(m - \tfrac{M}{2}\right)^2 + \tfrac{M^2}{2}, \tag{66}$$

so by (8)

$$E^{(0)} = \frac{\hbar^2 M^2}{4I} + \frac{\hbar^2}{I}\,k^2, \qquad k \equiv m - \frac{M}{2} = \frac{n}{2}. \tag{67}$$

The coupling connects m with $m \pm 1$, i.e. k with $k \pm 1$, with strength $\lambda/2$. Hence

$$H\big|_{\mathcal{H}_M} = \frac{\hbar^2 M^2}{4I} + h, \qquad h_{kk'} = \frac{\hbar^2}{I}k^2\,\delta_{kk'} + \frac{\lambda}{2}\left(\delta_{k',k+1} + \delta_{k',k-1}\right), \tag{68}$$

a real symmetric tridiagonal matrix. The label k runs over $\mathbb{Z}$ if M is even and over $\mathbb{Z} + \frac{1}{2}$ if M is odd. Consequently every even-M sector is identical to $M = 0$ and every odd-M sector to $M = 1$, up to the constant $\hbar^2 M^2/4I$; we verify this numerically in Table B.1.

6.2. Identification with the pendulum

In sector M the wave function factorises in the variables (18):

$$e^{im_1\theta_1 + im_2\theta_2} = e^{i(m_1+m_2)\Theta}\,e^{i(m_1-m_2)\varphi/2} = e^{iM\Theta}\,e^{ik\varphi}, \tag{69}$$

so $\Psi(\theta_1,\theta_2) = e^{iM\Theta}u(\varphi)/\sqrt{2\pi}$ with $u(\varphi) = \sum_k c_k\,e^{ik\varphi}/\sqrt{2\pi}$. For even M, k is an integer and $u(\varphi + 2\pi) = u(\varphi)$; for odd M, k is a half-integer and $u(\varphi + 2\pi) = -u(\varphi)$. Now consider

$$h = -\frac{\hbar^2}{I}\frac{\mathrm{d}^2}{\mathrm{d}\varphi^2} + \lambda\cos\varphi. \tag{70}$$

Acting on $\chi_k(\varphi) = e^{ik\varphi}/\sqrt{2\pi}$, the derivative term gives $(\hbar^2/I)k^2\chi_k$ and $\lambda\cos\varphi\,\chi_k = \frac{\lambda}{2}(\chi_{k+1} + \chi_{k-1})$, so the matrix of (70) in the basis χ_k is exactly (68). This is the relative Hamiltonian H_{rel} of (20), quantised with $p = -i\hbar\,\partial_\varphi$: *each sector of the two-rotor problem is exactly the quantum pendulum*, with periodic boundary conditions for even M and antiperiodic ones for odd M.

6.3. Reduction to the standard Mathieu equation

The eigenvalue equation $hu = E_{\mathrm{rel}}u$ reads $-(\hbar^2/I)u'' + \lambda\cos\varphi\,u = E_{\mathrm{rel}}u$. Substituting $\varphi = 2z$, so that $\mathrm{d}^2/\mathrm{d}\varphi^2 = \frac{1}{4}\mathrm{d}^2/\mathrm{d}z^2$, and multiplying by $-4I/\hbar^2$,

$$\frac{\mathrm{d}^2u}{\mathrm{d}z^2} + (a - 2q\cos 2z)\,u = 0, \qquad \boxed{a = \frac{4IE_{\mathrm{rel}}}{\hbar^2}, \qquad q = \frac{2\lambda I}{\hbar^2} = 4\alpha} \tag{71}$$

the Mathieu equation in standard form [11, 13]. A 2π-periodic $u(\varphi)$ is π-periodic in z; the corresponding solutions are $\mathrm{ce}_{2r}(z,q)$ (even) and $\mathrm{se}_{2r+2}(z,q)$ (odd), with characteristic values $a_{2r}(q)$ and $b_{2r+2}(q)$. An antiperiodic $u(\varphi)$ is π-antiperiodic in z; the solutions are ce_{2r+1} and se_{2r+1}, with $a_{2r+1}(q)$, $b_{2r+1}(q)$. Therefore

$$E = \frac{\hbar^2 M^2}{4I} + \frac{\hbar^2}{4I} \times \begin{cases} a_0 < b_2 < a_2 < b_4 < \dots, & M \text{ even}, \\ b_1 < a_1 < b_3 < a_3 < \dots, & M \text{ odd}, \end{cases} \tag{72}$$

which we have checked to all printed digits against exact diagonalisation (Table B.1). The orderings follow from the interlacing $a_0 < b_1 < a_1 < b_2 < a_2 < \dots$ valid for $q > 0$; for odd M the lowest level is b_1, whose eigenfunction $\mathrm{se}_1 \sim \sin z$ is largest at the potential minimum $z = \pi/2$. The separatrix $E_{\rm rel} = \lambda$ is at $a = 2q$. Parity $z \to -z$ is the exchange P of Sec. 2.6: ce and se are its even and odd eigenfunctions.

The number of librating levels follows from the Bohr–Sommerfeld rule: the area enclosed by the separatrix $p = \pm\sqrt{I\lambda(1-\cos\varphi)} = \pm\sqrt{2I\lambda}\,|\sin(\varphi/2)|$ is $\oint p\,\mathrm{d}\varphi = 2\sqrt{2I\lambda}\int_0^{2\pi}\sin(\varphi/2)\mathrm{d}\varphi = 8\sqrt{2I\lambda}$, and dividing by $2\pi\hbar$,

$$N_{\rm lib} \simeq \frac{8\sqrt{2I\lambda}}{2\pi\hbar} = \frac{4}{\pi}\sqrt{q}. \tag{73}$$

For $\lambda I/\hbar^2 = 50, 200, 800, 3200$ this gives $12.7, 25.5, 50.9, 101.9$, against $13, 25, 51, 102$ counted.

6.4. The entanglement entropy is a Shannon entropy

For an eigenstate $\Psi = e^{iM\Theta}u(\varphi)/\sqrt{2\pi}$ with $u = \sum_k c_k\chi_k$, the state is

$$|\Psi\rangle = \sum_k c_k \left|m_1 = \tfrac{M}{2}+k,\ m_2 = \tfrac{M}{2}-k\right\rangle. \tag{74}$$

Each value of m_1 appears with a single m_2, so by (52)

$$\rho_1 = \sum_k |c_k|^2 \left|\tfrac{M}{2}+k\right\rangle\!\left\langle\tfrac{M}{2}+k\right|, \qquad \boxed{S = -\sum_k |c_k|^2 \ln|c_k|^2} \tag{75}$$

exactly, to all orders in λ. *The entanglement entropy of the two rotors is the Shannon entropy of the Fourier coefficients of the Mathieu function.* The Schmidt basis is fixed by angular-momentum conservation, and the entropy depends on M only through its parity (and through $E_{\rm rel}$). The perturbative result (53) is the small-q expansion of (75).

6.5. Parity sectors in the plane-wave basis

Above the separatrix the levels of (70) form exponentially close doublets (counter-rotating states), so a dense eigensolver applied to (68) returns arbitrary mixtures of the two members. We diagonalise in the eigenspaces of $P : k \to -k$ separately. For even M the even sector has basis $\{\chi_0, (\chi_k + \chi_{-k})/\sqrt{2}\}_{k\ge1}$, in which

$$\langle\chi_0|\,h\,\tfrac{1}{\sqrt2}(\chi_1+\chi_{-1}) = \tfrac{1}{\sqrt2}\left(\tfrac{\lambda}{2}+\tfrac{\lambda}{2}\right) = \tfrac{\lambda}{\sqrt2}, \tag{76}$$

while all other off-diagonal elements are $\lambda/2$; the odd sector has basis $\{(\chi_k-\chi_{-k})/\sqrt{2}\}_{k\ge1}$ with off-diagonal elements $\lambda/2$. For odd M the even and odd combinations of $\chi_{\pm1/2}$ acquire diagonal elements $\hbar^2/4I \pm \lambda/2$: the coupling $\chi_{1/2} \leftrightarrow \chi_{-1/2}$ splits the $n = \pm1$ pair at first order, which is the resonance of Sec. 3 seen from the exact side.

7. Classical entanglement entropy

7.1. The microcanonical distribution

The classical Hamiltonian is (3) with real p_i. It is integrable, with two independent integrals in involution, E and J. Following [7], the classical counterpart of an eigenstate with definite

energy and total angular momentum is the microcanonical distribution on the corresponding level set,

$$W(\theta_1, p_1; \theta_2, p_2) = \mathcal{C}\ \delta(E - H)\ \delta(J - p_1 - p_2)\,, \tag{77}$$

normalised by $\int \mathrm{d}\theta_1 \mathrm{d}p_1 \mathrm{d}\theta_2 \mathrm{d}p_2\, W = 1$. By Liouville's theorem this measure is invariant under the flow; on each invariant torus it is uniform in the angle variables, which is the same as the time average along a generic orbit. One point deserves attention. For $E_{\mathrm{rel}} > \lambda$ the level set consists of *two* tori, $p > 0$ and $p < 0$ (the two senses of relative rotation), and (77) weights them equally; for $|E_{\mathrm{rel}}| < \lambda$ there is one torus. This matches the quantum side, where the exact eigenstates are parity eigenstates, symmetric under $p \to -p$.

7.2. Integrating out the second rotor

The reduced distribution of rotor 1 is $W(\theta_1, p_1) = \int \mathrm{d}\theta_2\, \mathrm{d}p_2\, W$. The second delta function fixes $p_2 = J - p_1$:

$$W(\theta_1, p_1) = \mathcal{C} \int_0^{2\pi} \mathrm{d}\theta_2\ \delta(E - K(p_1) - \lambda \cos(\theta_1 - \theta_2))\,, \qquad K(p_1) = \frac{p_1^2 + (J - p_1)^2}{2I}. \tag{78}$$

With $p_1 = J/2 + p$ the identity $p_1^2 + (J - p_1)^2 = J^2/2 + 2p^2$ gives $E - K = E_{\mathrm{rel}} - p^2/I$. Changing variable to $\varphi = \theta_1 - \theta_2$ ($|\mathrm{d}\varphi| = |\mathrm{d}\theta_2|$) and using periodicity to shift the interval of integration to a full period,

$$W(\theta_1, p_1) = \mathcal{C} \int_0^{2\pi} \mathrm{d}\varphi\ \delta(g(\varphi))\,, \qquad g(\varphi) = E_{\mathrm{rel}} - \frac{p^2}{I} - \lambda \cos\varphi. \tag{79}$$

For a function with simple zeros φ_j, $\delta(g(\varphi)) = \sum_j \delta(\varphi - \varphi_j)/|g'(\varphi_j)|$. Here $g'(\varphi) = \lambda \sin\varphi$, and $g(\varphi_j) = 0$ means $\cos\varphi_j = \xi \equiv (E_{\mathrm{rel}} - p^2/I)/\lambda$. For $|\xi| < 1$ there are two zeros in $[0, 2\pi)$, $\varphi = \pm \arccos\xi$, each with $|g'| = \lambda\sqrt{1 - \xi^2} = \sqrt{\lambda^2 - (E_{\mathrm{rel}} - p^2/I)^2}$; for $|\xi| > 1$ there are none. Hence

$$\boxed{W(\theta_1, p_1) = \frac{2\mathcal{C}}{\sqrt{\lambda^2 - \left(E_{\mathrm{rel}} - p^2/I\right)^2}}\ \Theta\big(\lambda - \big|E_{\mathrm{rel}} - p^2/I\big|\big)\,, \qquad p = p_1 - \frac{J}{2}} \tag{80}$$

independent of θ_1, as it must be: no absolute angle is distinguished. The classical correlation is carried entirely by the momentum. Rotor 1 is constrained by the requirement that the interaction energy $E_{\mathrm{rel}} - p^2/I$ lie in the range $[-\lambda, \lambda]$ accessible to $\lambda\cos\varphi$, and this constraint, inherited from the unobserved rotor, spreads p_1 over a band.

7.3. The support and its cusp at the separatrix

The band $|E_{\mathrm{rel}} - p^2/I| \le \lambda$ is

$$\begin{aligned} &|p| \le \sqrt{I(E_{\mathrm{rel}} + \lambda)}, && -\lambda < E_{\mathrm{rel}} < \lambda \quad \text{(libration)},\\ &\sqrt{I(E_{\mathrm{rel}} - \lambda)} \le |p| \le \sqrt{I(E_{\mathrm{rel}} + \lambda)}, && E_{\mathrm{rel}} > \lambda \quad \text{(rotation)}, \end{aligned} \tag{81}$$

with total length

$$L(E_{\mathrm{rel}}) = \begin{cases} 2\sqrt{I(E_{\mathrm{rel}} + \lambda)}, & E_{\mathrm{rel}} < \lambda,\\ 2\left[\sqrt{I(E_{\mathrm{rel}} + \lambda)} - \sqrt{I(E_{\mathrm{rel}} - \lambda)}\right], & E_{\mathrm{rel}} > \lambda. \end{cases} \tag{82}$$

L increases through the libration regime, reaches its maximum $2\sqrt{2I\lambda}$ at the separatrix, and decreases above it, with $\mathrm{d}L/\mathrm{d}E_{\mathrm{rel}} \to -\infty$ as $E_{\mathrm{rel}} \to \lambda^+$ because of the $-\sqrt{E_{\mathrm{rel}} - \lambda}$ term. The support therefore has a cusp maximum at the separatrix. Since the entropy of a distribution on a set of length L is at most $\ln(L/\hbar)$, this is the origin of the entropy maximum found in Sec. 9.

7.4. Normalisation and the classical period

Integrating (80) over θ_1 gives the momentum density $w(p_1) = 2\pi W$. Its normalisation requires the band integral

$$B(E_{\mathrm{rel}}) \equiv \int_{\text{band, p>0}} \frac{\mathrm{d}p}{\sqrt{\lambda^2 - (E_{\mathrm{rel}} - p^2/I)^2}}. \tag{83}$$

Substitute $E_{\mathrm{rel}} - p^2/I = \lambda\cos u$: then $p = \sqrt{I(E_{\mathrm{rel}} - \lambda\cos u)}$, $\mathrm{d}p = I\lambda\sin u\,\mathrm{d}u/(2p)$ and $\sqrt{\lambda^2 - \lambda^2\cos^2 u} = \lambda\sin u$, so

$$B = \frac{I}{2}\int \frac{\mathrm{d}u}{p(u)} = \frac{\sqrt{I}}{2}\int \frac{\mathrm{d}u}{\sqrt{E_{\mathrm{rel}} - \lambda\cos u}}. \tag{84}$$

Rotation ($E_{\mathrm{rel}} > \lambda$): $u \in [0, \pi]$. With $u = \pi - 2t$, $\cos u = -\cos 2t = -1 + 2\sin^2 t$ and $E_{\mathrm{rel}} - \lambda\cos u = (E_{\mathrm{rel}} + \lambda)(1 - k^2\sin^2 t)$ with $k^2 = 2\lambda/(E_{\mathrm{rel}} + \lambda) < 1$, whence

$$B = \frac{\sqrt{I}}{2}\cdot\frac{2}{\sqrt{E_{\mathrm{rel}} + \lambda}}\int_0^{\pi/2}\frac{\mathrm{d}t}{\sqrt{1 - k^2\sin^2 t}} = \sqrt{\frac{I}{E_{\mathrm{rel}} + \lambda}}\,\mathsf{K}(k), \qquad k^2 = \frac{2\lambda}{E_{\mathrm{rel}} + \lambda}. \tag{85}$$

Libration ($|E_{\mathrm{rel}}| < \lambda$): $p^2 \geq 0$ restricts u to $[u_0, \pi]$ with $\cos u_0 = E_{\mathrm{rel}}/\lambda$. With $v = \pi - u$ and $\sin(v/2) = \kappa\sin t$, $\kappa^2 = (E_{\mathrm{rel}} + \lambda)/2\lambda < 1$, one finds $E_{\mathrm{rel}} + \lambda\cos v = (E_{\mathrm{rel}} + \lambda)\cos^2 t$ and $\mathrm{d}v = 2\kappa\cos t\,\mathrm{d}t/\sqrt{1 - \kappa^2\sin^2 t}$, so that

$$B = \frac{\sqrt{I}}{2}\cdot\frac{2\kappa}{\sqrt{E_{\mathrm{rel}} + \lambda}}\int_0^{\pi/2}\frac{\mathrm{d}t}{\sqrt{1 - \kappa^2\sin^2 t}} = \sqrt{\frac{I}{2\lambda}}\,\mathsf{K}(\kappa), \qquad \kappa^2 = \frac{E_{\mathrm{rel}} + \lambda}{2\lambda}. \tag{86}$$

Here K is the complete elliptic integral of the first kind. Both forms agree with direct quadrature to eight digits (Appendix B). In either regime the momentum density has total weight $2B$ over all p (two bands, or one band symmetric in p), so

$$\mathcal{C} = \frac{1}{8\pi B}, \qquad w(p_1) = \frac{1}{2B}\,\frac{1}{\sqrt{\lambda^2 - (E_{\mathrm{rel}} - p^2/I)^2}}. \tag{87}$$

The band integral is, up to a constant, the classical period: with $\dot\varphi = \partial H_{\mathrm{rel}}/\partial p = 2p/I$, $\mathrm{d}t = I\mathrm{d}\varphi/2p$, and the same substitution shows that the period of rotation is $T = 2B$. At the separatrix $k, \kappa \to 1$ and $\mathsf{K} \simeq \ln\left(4/\sqrt{1 - k^2}\right)$ diverges logarithmically: the orbit lingers near the unstable point $\varphi = 0$, where $p \to 0$.

7.5. Boltzmann entropy and the elementary cell

The Boltzmann entropy of the reduced distribution is

$$S_{\rm cl} = -\int \mathrm{d}\theta_1\, \mathrm{d}p_1\; W \ln\left(W\Delta\right), \tag{88}$$

with Δ an elementary area making $W\Delta$ dimensionless. On a compact configuration space Δ is not free. Single-valuedness quantises $p_1 = \hbar m_1$, so each quantum state occupies the strip $\Delta\theta_1 \times \Delta p_1 = 2\pi \times \hbar = h$, and $\Delta = h$. Because $W = w(p_1)/2\pi$ is independent of θ_1, (88) reduces to

$$S_{\rm cl} = -\int \mathrm{d}p_1\; w \ln\left(w\hbar\right), \tag{89}$$

a differential entropy in units of $\hbar$. The object with the same status as the quantum entropy (75) — a Shannon entropy over the integers — is obtained by integrating w over the unit cells:

$$w_m = \int_{\hbar(m-1/2)}^{\hbar(m+1/2)} w(p_1)\,\mathrm{d}p_1, \qquad \sum_m w_m = 1, \qquad S_{\rm cl}^{\rm bin} = -\sum_m w_m \ln w_m. \tag{90}$$

The comparison between (90) and (75) contains no adjustable constant. The two classical expressions are related as follows. On cell m let $\varrho(p) = w(p)/w_m$, a normalised density on an interval of length $\hbar$. Then

$$S_{\rm cl} = -\sum_m w_m \int_{\rm cell} \varrho \ln\left(w_m \varrho \hbar\right) \mathrm{d}p = S_{\rm cl}^{\rm bin} - \sum_m w_m\, D_{\rm KL}\big(\varrho \,\|\, \hbar^{-1}\big)\,, \tag{91}$$

where $D_{\rm KL}(\varrho\|\hbar^{-1}) = \int \varrho \ln(\varrho\hbar)\,\mathrm{d}p \geq 0$ is the relative entropy of ϱ with respect to the uniform density on the cell. Hence $S_{\rm cl}^{\rm bin} \geq S_{\rm cl}$, with equality when w is constant on each cell: coarse-graining raises the entropy. The difference comes mainly from the cells containing the inverse-square-root singularities at the band edges, whose weight decreases as (band width in cells)$^{-1/2}$; for the arcsine law of Sec. 8 we find $S_{\rm cl}^{\rm bin} - S_{\rm cl} = 0.20, 0.064, 0.020$ for $\gamma = 10, 100, 1000$, decreasing as $\gamma^{-1/2}$.

8. Deep rotation: an exactly solvable limit and the constant $1 - \ln 2$

8.1. Reduction to a Wannier–Stark ladder

Consider a level with large relative angular momentum, $k_0 = n/2 \gg 1$, and write $k = k_0 + j$. Then

$$\frac{\hbar^2}{I}k^2 = \frac{\hbar^2}{I}k_0^2 + F\,j + \frac{\hbar^2}{I}j^2, \qquad F \equiv \frac{2\hbar^2 k_0}{I} = \frac{\hbar^2 n}{I}. \tag{92}$$

Over the range of j that will turn out to matter, $|j| \lesssim \gamma$ with γ defined below, the quadratic term is smaller than the linear one by $|j|/n \leq \gamma/n$. Dropping it, (68) becomes, up to the constant $\hbar^2 k_0^2/I$,

$$h_{\rm WS}\, c_j = F\, j\, c_j + \frac{\lambda}{2}\left(c_{j+1} + c_{j-1}\right), \tag{93}$$

a particle hopping on a lattice in a uniform field: the Wannier–Stark problem. Its eigenvectors follow from the Bessel recurrence $J_{\nu-1}(x) + J_{\nu+1}(x) = (2\nu/x)J_\nu(x)$. Try $c_j = J_{j-s}(x)$ with $s \in \mathbb{Z}$:

$$\frac{\lambda}{2}\left(c_{j+1} + c_{j-1}\right) = \frac{\lambda}{2}\cdot\frac{2(j-s)}{x}J_{j-s}(x) = \frac{\lambda(j-s)}{x}c_j, \tag{94}$$

so $h_{\mathrm{WS}}c_j = [Fj + \lambda(j-s)/x]\,c_j$. This is independent of j, as an eigenvalue must be, if and only if $x = -\lambda/F$; the eigenvalue is then Fs. Using $J_\nu(-x) = (-1)^\nu J_\nu(x)$, the eigenvector centred on s has

$$|c_j|^2 = J_{j-s}^2(\gamma), \qquad \boxed{\gamma \equiv \frac{\lambda}{F} = \frac{\lambda I}{\hbar^2 n} = \frac{2\alpha}{n}} \tag{95}$$

Normalisation is the Neumann identity $\sum_j J_j^2(\gamma) = 1$. The Wannier–Stark levels Fs are equally spaced and non-degenerate, and the eigenvectors are translates of one another, so all have the same entropy.

Validity. The neglected term is small when $\gamma \ll n$, i.e. $\lambda I/\hbar^2 \ll n^2$; with $E_{\mathrm{rel}} \simeq \hbar^2 n^2/4I$ this is $\gamma/n = \lambda/4E_{\mathrm{rel}} \ll 1$, deep in the rotation regime. Exact diagonalisation confirms (95) pointwise to 3×10^{-3} at $\gamma = 2,5$ ($\gamma/n = 0.01$), and the entropy to four or five digits for $n = 100$–400 (Fig. 2a).

8.2. The entanglement entropy as a function of one variable

From (75) and (95), the branch entropy in deep rotation is a universal function of γ alone:

$$\boxed{S_{\mathrm{branch}}(\gamma) = -\sum_{j\in\mathbb{Z}} J_j^2(\gamma) \ln J_j^2(\gamma)} \tag{96}$$

(for the exact parity eigenstate add $\ln 2$, Sec. 5.7). Its two limits are the perturbative and the classical regimes.

Small γ. From $J_0(\gamma) = 1 - \gamma^2/4 + \ldots$ and $J_{\pm1}(\gamma) = \pm\gamma/2 + \ldots$, the weights are $1 - \gamma^2/2$ and $\gamma^2/4$ (twice), so

$$S_{\mathrm{branch}} \simeq 2 \cdot \frac{\gamma^2}{4} \ln \frac{4}{\gamma^2} + \frac{\gamma^2}{2} = \gamma^2 \left[\ln \frac{2}{\gamma} + \frac{1}{2} \right]. \tag{97}$$

Since $\gamma^2/4 = \alpha^2/n^2 \simeq \varepsilon_\pm$, this is exactly the large-n perturbative result (60), including the factor 2 in the logarithm and the normalisation term $\frac{1}{2}$.

Large γ. Here the weights become many and the classical description applies; we treat this next.

8.3. The classical counterpart: the arcsine law

In the same limit the classical density (80) simplifies. Near $p_0 = \sqrt{IE_{\mathrm{rel}}}$, linearise $p^2/I \simeq p_0^2/I + 2p_0(p-p_0)/I$, so that $E_{\mathrm{rel}} - p^2/I \simeq -2p_0(p-p_0)/I$ and

$$w \propto \frac{1}{\sqrt{\lambda^2 - 4p_0^2(p-p_0)^2/I^2}} \propto \frac{1}{\sqrt{a^2 - (p-p_0)^2}}, \qquad a = \frac{I\lambda}{2p_0}. \tag{98}$$

In units of $\hbar$, with $p_0 = \hbar n/2$, the half-width is $a/\hbar = \lambda I/\hbar^2 n = \gamma$. The classical weights are the arcsine law on $|j| < \gamma$,

$$w(j) = \frac{1}{\pi\sqrt{\gamma^2 - j^2}}, \tag{99}$$

and its differential entropy is (Appendix A)

$$S_{\mathrm{cl}} = -\int_{-\gamma}^{\gamma} w \ln w \,\mathrm{d}j = \ln \frac{\pi\gamma}{2}. \tag{100}$$

For the parity-symmetric distribution, which consists of two such bands of equal weight at $\pm p_0$, one adds $\ln 2$. Using $\hbar n = 2\sqrt{IE_{\rm rel}}$, $\gamma = \lambda I/\hbar^2 n = (\lambda/2\hbar)\sqrt{I/E_{\rm rel}}$, so

$$S_{\rm cl}^{\rm branch} = \ln\frac{\pi\gamma}{2}, \qquad S_{\rm cl}^{\rm two\ bands} = \ln(\pi\gamma) = \ln\left[\frac{\pi\lambda}{2\hbar}\sqrt{\frac{I}{E_{\rm rel}}}\right]. \tag{101}$$

This is the rotor counterpart of the DKS result (2): the entropy grows logarithmically with the coupling and decreases logarithmically with the energy of the relative motion, and it becomes negative — the classical description fails — when $\gamma < 2/\pi$, i.e. when the band covers less than about one cell.

8.4. Interference fringes in the quantum weights

The quantum weights are not the arcsine law: they oscillate about it. The Debye expansion of the Bessel function [12], written with $j = \gamma\cos\beta$, $0 < \beta < \pi$, reads

$$J_j(\gamma) \simeq \sqrt{\frac{2}{\pi\gamma\sin\beta}}\,\cos\left(\gamma\sin\beta - j\beta - \frac{\pi}{4}\right). \tag{102}$$

Squaring, with $\gamma\sin\beta = \sqrt{\gamma^2 - j^2}$ and $\cos^2 x = \frac{1}{2}(1+\cos 2x)$,

$$J_j^2(\gamma) \simeq w(j)\,[1+\cos\Phi_j], \qquad \Phi_j = 2\sqrt{\gamma^2-j^2} - 2j\arccos\frac{j}{\gamma} - \frac{\pi}{2}, \tag{103}$$

with $w(j)$ the arcsine law (99). The quantum weights are the classical ones modulated by fringes of *full* contrast. Their origin is the interference of the two classical relative angles $\varphi = \pm\arccos\xi$ that correspond to the same momentum. The local fringe wavenumber is

$$\frac{{\rm d}\Phi_j}{{\rm d}j} = -\frac{2j}{\sqrt{\gamma^2-j^2}} - 2\arccos\frac{j}{\gamma} + \frac{2j}{\sqrt{\gamma^2-j^2}} = -2\beta, \tag{104}$$

which ranges over $(-2\pi, 0)$ across the band.

The same structure holds at any energy. In the k representation the eigenvalue equation $\frac{\hbar^2}{I}k^2 c_k + \frac{\lambda}{2}(c_{k+1}+c_{k-1}) = E_{\rm rel}c_k$ is a difference equation, and the WKB ansatz $c_k = A_k e^{i\chi_k}$ with slowly varying A_k and local wavenumber $\phi_k = \chi_{k+1} - \chi_k$ gives at leading order $E_{\rm rel} - \hbar^2k^2/I = \lambda\cos\phi_k$, which is the classical relation between the relative momentum $\hbar k$ and the relative angle. The two roots $\pm\phi_k$ give two solutions. The Casoratian $\mathcal{W}_k = \frac{\lambda}{2}(c_k\bar{c}_{k+1} - c_{k+1}\bar{c}_k)$ of any two solutions is independent of k, which for a single WKB wave gives $|A_k|^2\lambda\sin\phi_k = $ const, i.e. $|A_k|^2 \propto 1/\sqrt{\lambda^2 - (E_{\rm rel} - \hbar^2k^2/I)^2}$: the classical density (80). A real eigenvector is a superposition of the two waves with equal amplitude, so in general

$$|c_k|^2 \simeq w_k\,(1+\cos\Phi_k), \qquad \Phi_{k+1} - \Phi_k = 2\phi_k. \tag{105}$$

The classical distribution is the fringe average of the quantum one.

8.5. The entropy deficit

Insert $p_k = w_k(1+\cos\Phi_k)$ into $S_q = -\sum p_k \ln p_k$:

$$\begin{aligned} S_q &= -\sum_k w_k(1+\cos\Phi_k)\,[\ln w_k + \ln(1+\cos\Phi_k)] \\ &= S_{\rm cl}^{\rm bin} - \sum_k w_k \cos\Phi_k \ln w_k - \sum_k w_k(1+\cos\Phi_k)\ln(1+\cos\Phi_k). \end{aligned} \tag{106}$$

In the second term $\cos\Phi_k$ oscillates on the scale of a few cells while $w_k \ln w_k$ varies on the scale of the band, so the sum vanishes as the band widens (the discrete analogue of the Riemann–Lebesgue lemma). In the third, the phases $\Phi_k \bmod 2\pi$ are equidistributed wherever the increment $2\phi_k$ is not close to a multiple of 2π, and the sum becomes a phase average:

$$S_{\rm cl} - S_q \;\longrightarrow\; \frac{1}{2\pi}\int_0^{2\pi} (1+\cos\Phi)\ln(1+\cos\Phi)\,{\rm d}\Phi = 1 - \ln 2. \tag{107}$$

The integral is evaluated in Appendix A. Thus

$$\boxed{S_q = S_{\rm cl} - (1-\ln 2) + o(1)} \qquad 1-\ln 2 = 0.30685\ldots \tag{108}$$

in the semiclassical limit. The limiting deficit is universal: it does not depend on λ, n, M, or the energy, only on the full contrast of the two-branch interference.

Rate of approach. Equidistribution fails near the band edges, where $\phi_k \to 0$ or π and the fringes become stationary (the caustics of the momentum-space WKB, where it is replaced by an Airy function). Near the edge $j = \gamma - \delta$ one has $\beta \simeq \sqrt{2\delta/\gamma}$, so the phase accumulated over the last δ cells is $\int 2\beta\,{\rm d}j \simeq \frac{4\sqrt{2}}{3}\delta^{3/2}/\sqrt{\gamma}$. This reaches 2π at $\delta \sim \gamma^{1/3}$, and the arcsine weight within that distance of the edge is $\pi^{-1}\arccos(1-\delta/\gamma) \simeq \pi^{-1}\sqrt{2\delta/\gamma} \sim \gamma^{-1/3}$. Hence

$$S_{\rm cl} - S_q = 1 - \ln 2 - O\big(\gamma^{-1/3}\big)\,. \tag{109}$$

8.6. Numerical confirmation

Because (95) is explicit, (108) can be tested to very large γ by summing J_j^2 directly. Figure 2(b) shows $S_{\rm cl}^{\rm bin} - S_q$ for $\gamma = 10$ to 10^6. It rises monotonically, 0.233 at $\gamma = 10^3$, 0.270 at 10^4, 0.288 at 10^5 and 0.298 at 10^6, and the distance to $1-\ln 2$ halves per decade, consistent with $\gamma^{-1/3}$ (a factor 2.15 per decade).

8.7. Interpretation

The oscillator pair of [7] and the rotor pair differ in one structural respect. In the oscillator case no conservation law singles out a Schmidt basis; the reduced density matrix found in [7] is doubly degenerate, with eigenvalues following the smooth arcsine law and with the two-branch interference absorbed into the eigenvectors. For the rotors, conservation of J forces the Schmidt basis to be the angular-momentum basis (Sec. 6), so the interference cannot be absorbed into the eigenvectors and appears in the eigenvalues themselves, as the fringes (105). The leading logarithms of the quantum and classical entropies agree, as in the oscillator case; the constant term does not, and the difference is $1-\ln 2$. Equivalently, the classical formula (89) reproduces the quantum entropy if the cell is taken to be $\Delta = eh/2$ rather than h, since $\ln(e/2) = 1-\ln 2$.

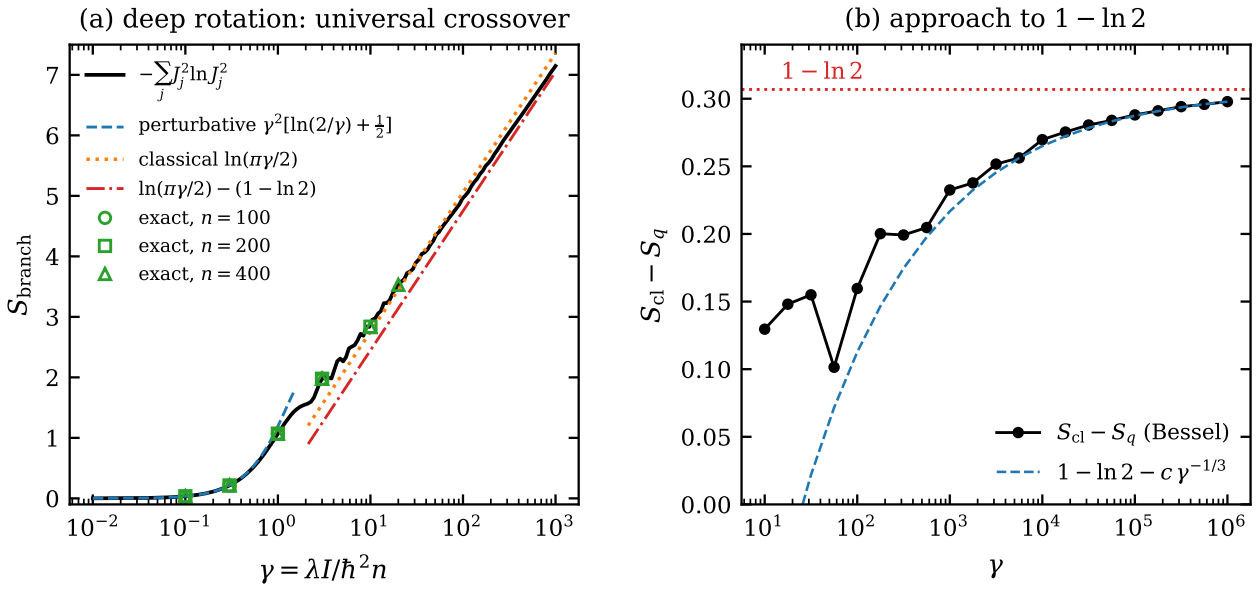


Figure 2: Deep rotation. (a) The branch entropy as a universal function of $\gamma = \lambda I/\hbar^2 n$: the Bessel result (96) (black), its perturbative limit (blue dashed), the classical arcsine entropy (orange dotted) and the classical entropy minus $1-\ln 2$ (red dash-dotted). Open symbols: exact diagonalisation at $n = 100, 200, 400$. (b) The deficit $S_{\rm cl}^{\rm bin} - S_q$ from the Bessel weights, approaching $1-\ln 2$ as $\gamma^{-1/3}$.

9. Generic energies and the separatrix

Away from deep rotation we compare (75), computed by exact diagonalisation in the parity sectors, with (90), computed by adaptive quadrature of (80) over each cell. Figure 3 shows both for $\lambda I/\hbar^2 = 50, 200, 800$ as a function of $E_{\rm rel}/\lambda$.

Agreement and deficit. The curves track each other over the whole range. The quantum entropy lies below the classical one throughout libration, by about 0.2, except within a few levels of the bottom of the well, where the band covers only a handful of cells and the classical description fails; above the separatrix it lies below by a smaller amount that grows with λ. Relative to $S \approx 3$–4 this is a few percent. Figure 4(a) shows the mechanism: at $\lambda I/\hbar^2 = 800$, $E_{\rm rel}/\lambda = 1.54$ the quantum weights oscillate about the classical envelope, the total-variation distance between the two distributions is 0.33, yet the entropies differ by less than 1%. Figure 4(b) shows the deficit averaged over energy windows as λ grows from 200 to 51 200: in libration it rises from 0.18 to 0.24, in rotation from 0.01 to 0.15. Both increase monotonically towards $1-\ln 2$, but more slowly than in the pure Wannier–Stark case, because at these couplings the bands contain only 10–500 cells and the caustic regions carry a large share of the weight.

The separatrix. Both entropies are maximal near $E_{\rm rel} = \lambda$; for $\lambda I/\hbar^2 = 50$, 200 and 800 the quantum maximum is at $E_{\rm rel}/\lambda = 1.13$, 1.12 and 0.90. The classical reason is the cusp in the support (82): the band of accessible momenta is widest at the separatrix. On the quantum side this is where the non-degenerate librational levels give way to the rotational doublets. The deviation in Fig. 3(b) changes abruptly there, because the fringe structure changes: below the separatrix the two interfering branches are the two halves of one libration orbit, above it the two halves of one rotation orbit, and the band edges (caustics) move from $p = \pm\sqrt{I(E_{\rm rel}+\lambda)}$ alone

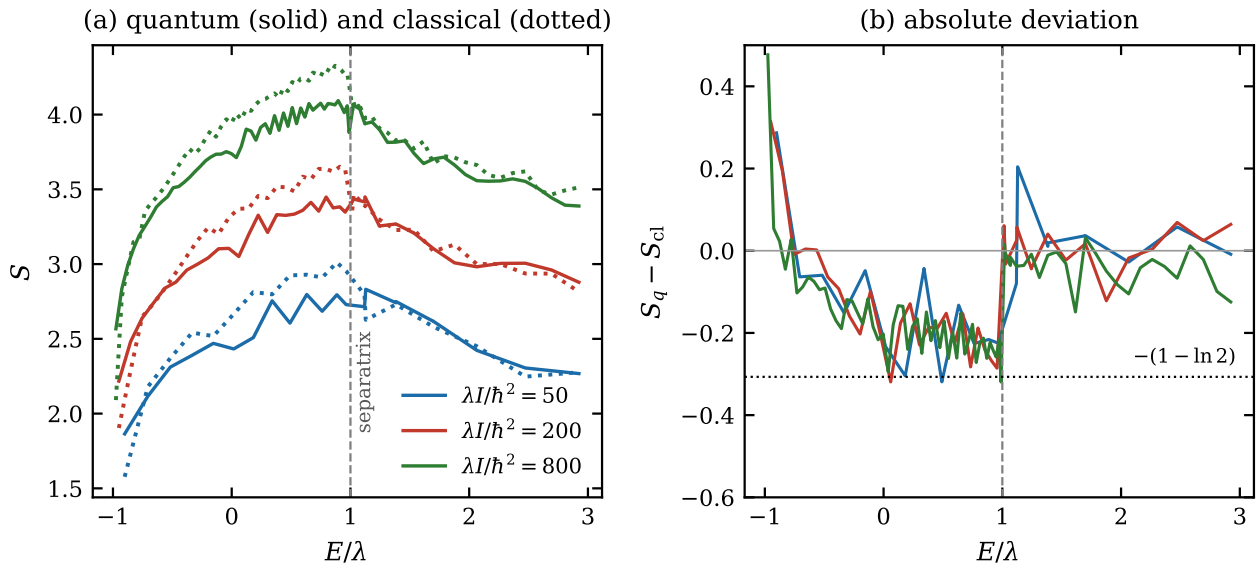


Figure 3: (a) Quantum entanglement entropy (solid) and classical coarse-grained entropy (dotted) as functions of $E_{\rm rel}/\lambda$ for $\lambda I/\hbar^2 = 50$, 200 and 800. The dashed line marks the separatrix. (b) Absolute deviation $S_q - S_{\rm cl}^{\rm bin}$, with the asymptotic value $-(1-\ln 2)$ of (108) shown dotted.

to include $p = \pm\sqrt{I(E_{\rm rel} - \lambda)}$. The separatrix of a classical system is the locus of an excited-state quantum phase transition [8, 9, 10], and the entropy maximum should be understood in that context.

10. Deep libration

10.1. Harmonic approximation

For $\lambda I/\hbar^2 \gg 1$ the ground state of h is localised near the minimum $\varphi = \pi$. Writing $x = \varphi - \pi$ and expanding $\lambda\cos\varphi = -\lambda\cos x = -\lambda + \frac{1}{2}\lambda x^2 - \frac{1}{24}\lambda x^4 + O(x^6)$,

$$h = -\lambda + \frac{p^2}{2\mu} + \frac{1}{2}\mu\omega^2 x^2 - g x^4 + \dots, \qquad \mu = \frac{I}{2}, \quad \mu\omega^2 = \lambda, \quad g = \frac{\lambda}{24}, \tag{110}$$

so $\omega = \sqrt{2\lambda/I}$. Neglecting g, the ground state is $u_0(x) \propto \exp(-\mu\omega x^2/2\hbar)$, with $\langle x^2\rangle = \hbar/2\mu\omega$ and $\langle p^2\rangle = \mu\hbar\omega/2$. Its Fourier coefficients are, up to the phase $e^{-ik\pi}$, a Gaussian in $k = p/\hbar$ with variance

$$\sigma_0^2 = \frac{\langle p^2\rangle}{\hbar^2} = \frac{\mu\omega}{2\hbar} = \frac{I}{4\hbar}\sqrt{\frac{2\lambda}{I}} = \frac{1}{4}\sqrt{\frac{2\lambda I}{\hbar^2}}. \tag{111}$$

The wrap-around of the Gaussian on the circle is negligible when $\langle x^2\rangle^{1/2} \ll \pi$, which holds here.

10.2. Entropy of a sampled Gaussian

For $p_k = e^{-k^2/2\sigma^2}/Z$ on the integers, Poisson summation gives

$$Z = \sum_{k\in\mathbb{Z}} e^{-k^2/2\sigma^2} = \sqrt{2\pi}\,\sigma\left[1 + 2\sum_{\ell\geq 1} e^{-2\pi^2\sigma^2\ell^2}\right], \qquad \sum_k k^2 p_k = \sigma^2\left[1 + O\Big(\sigma^2 e^{-2\pi^2\sigma^2}\Big)\right]. \tag{112}$$

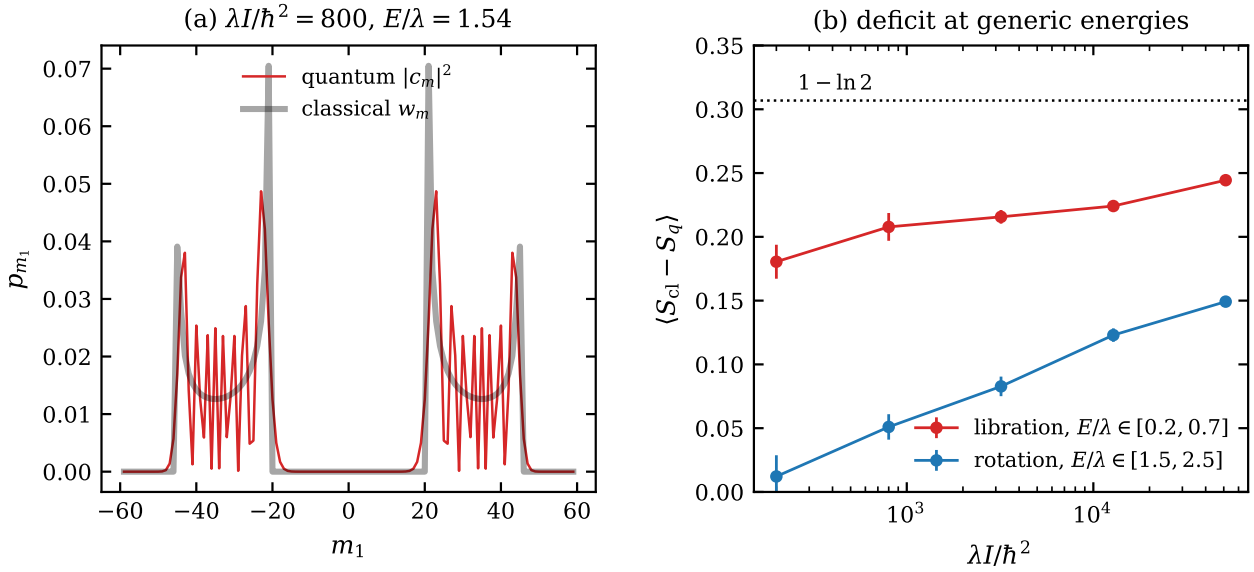


Figure 4: (a) Momentum distribution of rotor 1 at $\lambda I/\hbar^2 = 800$, $E_{\rm rel}/\lambda = 1.54$: the quantum weights (red) oscillate with full contrast about the classical cell weights (grey), as (105) predicts. (b) Deficit $\langle S_{\rm cl}^{\rm bin} - S_q \rangle$ averaged over energy windows in libration and in rotation, against λ; error bars are standard errors over the window.

Since $-\ln p_k = \ln Z + k^2/2\sigma^2$,

$$S = \ln Z + \frac{\langle k^2 \rangle}{2\sigma^2} = \frac{1}{2} \ln \left(2\pi e \sigma^2 \right) + O\Big(\sigma^2 e^{-2\pi^2 \sigma^2} \Big) . \tag{113}$$

The discreteness correction is exponentially small (below 10^{-8} already for $\sigma^2 \simeq 1$). With $\sigma = \sigma_0$,

$$S_0 = \frac{1}{2} \ln \left(2\pi e \cdot \frac{1}{4} \sqrt{\frac{2\lambda I}{\hbar^2}} \right) = \frac{1}{2} \ln \frac{\pi e}{2} + \frac{1}{4} \ln 2 + \frac{1}{4} \ln \frac{\lambda I}{\hbar^2} = \frac{1}{4} \ln \frac{\lambda I}{\hbar^2} + \frac{1}{2} \ln \frac{\pi e}{\sqrt{2}}, \tag{114}$$

where $\frac{1}{2} \ln(\pi e/2) + \frac{1}{4} \ln 2 = \frac{1}{2} \ln(\pi e \sqrt{2}/2)$ was used.

10.3. The leading anharmonic correction

The quartic term in (110) narrows the momentum distribution. To first order in g we need $\langle p^2 \rangle$. The virial theorem follows from $\langle [h, xp] \rangle = 0$ in an eigenstate: since $[p^2/2\mu, x]p = -i\hbar p^2/\mu$ and $x[V(x), p] = i\hbar x V'(x)$, it gives $2\langle T \rangle = \langle xV' \rangle$. With $V = V_2 - V_4$, $V_2 = \frac{1}{2}\mu\omega^2 x^2$, $V_4 = g x^4$, one has $xV' = 2V_2 - 4V_4$, so $\langle V_2 \rangle = \langle T \rangle + 2\langle V_4 \rangle$. Inserting this into $E = \langle T \rangle + \langle V_2 \rangle - \langle V_4 \rangle$ gives $\langle T \rangle = \frac{1}{2}(E - \langle V_4 \rangle)$. To first order, $E = \frac{1}{2}\hbar\omega - g\langle x^4 \rangle_0$ and $\langle V_4 \rangle = g\langle x^4 \rangle_0$, so

$$\langle T \rangle = \tfrac{1}{4}\hbar\omega - g\langle x^4 \rangle_0, \qquad \langle x^4 \rangle_0 = 3\langle x^2 \rangle_0^2 = \frac{3\hbar^2}{4\mu^2\omega^2}. \tag{115}$$

Therefore

$$\langle p^2 \rangle = 2\mu\langle T \rangle = \frac{\mu\hbar\omega}{2} - \frac{3g\hbar^2}{2\mu\omega^2} = \frac{\mu\hbar\omega}{2} - \frac{\hbar^2}{16}, \qquad \sigma^2 = \sigma_0^2 - \frac{1}{16}, \tag{116}$$

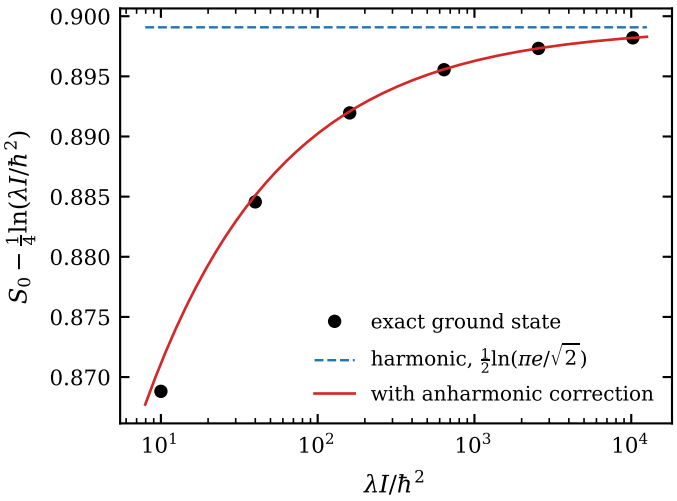


Figure 5: Deep libration: $S_0 - \frac{1}{4}\ln(\lambda I/\hbar^2)$ for the ground state from exact diagonalisation (points), the harmonic constant $\frac{1}{2}\ln(\pi e/\sqrt{2})$ (dashed), and (118) with the anharmonic correction (solid).

where $g/\mu\omega^2 = 1/24$ was used. The quartic term also makes the distribution non-Gaussian, but this does not change the entropy at first order. If $p = \phi_\sigma(1 + \epsilon\eta)$ with ϕ_σ the Gaussian of the exact variance, then η is orthogonal (with weight ϕ_σ) to 1, k and k^2, and

$$-\sum p \ln p = -\sum \phi_\sigma \ln \phi_\sigma - \epsilon \sum \phi_\sigma \eta \,(1 + \ln \phi_\sigma) + O(\epsilon^2) = S[\phi_\sigma] + O(\epsilon^2), \tag{117}$$

because $\ln \phi_\sigma$ is a quadratic polynomial in k. Hence $S = \frac{1}{2}\ln(2\pi e\sigma^2)$ to first order, and expanding $\frac{1}{2}\ln(\sigma_0^2 - \frac{1}{16}) = \frac{1}{2}\ln\sigma_0^2 - 1/32\sigma_0^2 + \ldots$,

$$\boxed{S_0 = \frac{1}{4}\ln\frac{\lambda I}{\hbar^2} + \frac{1}{2}\ln\frac{\pi e}{\sqrt{2}} - \frac{1}{8\sqrt{2\lambda I/\hbar^2}} + O\left(\frac{\hbar^2}{\lambda I}\right)} \tag{118}$$

with $\frac{1}{2}\ln(\pi e/\sqrt{2}) = 0.89908$. Figure 5 compares this with exact diagonalisation. For $\lambda I/\hbar^2 =$ 10, 40, 160, 640, 2560 and 10240 the residual after the harmonic terms is -0.03027, -0.01451, -0.00712, -0.00353, -0.00176 and -0.00088, and its ratio to the predicted correction is 1.083, 1.039, 1.019, 1.009, 1.005 and 1.002, converging to unity as $O(\lambda^{-1/2})$.

11. Discussion

We have carried the programme of [7] from two coupled oscillators to two coupled rotors. The central conclusion survives: in the semiclassical regime the entanglement entropy of an eigenstate is reproduced, at the level of its leading logarithm, by the Boltzmann entropy of the reduced classical phase-space distribution, and this holds for a coupling that is not bilinear and across a change in the topology of the classical orbits. The model is, however, rich enough to show where and how the correspondence stops.

The cell is fixed, and the equality is not exact. Compactness quantises the angular momentum, so the classical distribution is coarse-grained onto the integers that label the quantum Schmidt weights, and the comparison contains no free constant. With nothing to adjust, the

two entropies differ by $1 - \ln 2$ in the semiclassical limit. We derived this in deep rotation, where the quantum weights are exactly $J_j^2(\gamma)$ and the classical weights are the arcsine law, and confirmed it numerically to $\gamma = 10^6$; at generic energies the same deficit is approached more slowly. In the language of [7], the classical formula reproduces the quantum entropy only if the cell is taken as $eh/2$.

Why the oscillators and rotors differ. The constant arises from the interference of the two classical branches that correspond to one value of the observed momentum. For the rotors, conservation of total angular momentum fixes the Schmidt basis to be the angular-momentum basis, so the interference has nowhere to go but into the Schmidt weights, where it appears as full-contrast fringes. For the oscillators no conservation law fixes the Schmidt basis, the reduced density matrix is doubly degenerate, and the interference is carried by the eigenvectors. We expect the same distinction — whether a symmetry ties the Schmidt basis to the observable whose classical distribution is being compared — to decide the constant term in other systems.

Entropy and distribution are different statements. At $\lambda I/\hbar^2 = 800$ the quantum and classical momentum distributions have total-variation distance 0.33 while their entropies differ by less than one percent. The classical–quantum correspondence for entanglement entropy is a statement about one coarse functional of the state.

The separatrix. Both entropies are maximal there, because the support of the classical momentum distribution has a cusp maximum, and the deviation between them changes abruptly as the fringe structure changes. The separatrix is the classical locus of an excited-state quantum phase transition (ESQPT) [8, 9, 10]; entropic signatures of ESQPTs have been studied, and the entropy maximum reported here should be positioned with respect to that literature.

A point of technique. The normalisation constant of a perturbed state is of order λ^2 and may be dropped from a state kept to order λ. It may not be dropped from the density matrix at the same order, because ρ is bilinear in the amplitudes. Here it supplies the whole linear entropy and the non-logarithmic part of the von Neumann entropy; omitting it gives a trace and a purity larger than one and an entropy low by 10–14%. Checking $\operatorname{Tr}\rho_1 = 1$ and $S_L \geq 0$ at the order kept catches the omission.

Outlook. Three extensions are natural. A quench of λ across the separatrix can be followed with the same Mathieu machinery and would show how entanglement is generated dynamically. Coupling $N > 2$ rotors makes the relative motion non-integrable; whether the correspondence, and the constant $1 - \ln 2$, survive chaos is the question most relevant to the high-energy setting that motivated [7]. Finally, the Wannier–Stark structure of Sec. 8 suggests that the Bessel form of the Schmidt weights is generic for any pair of systems with a conserved total charge and a linear dispersion of the relative charge.

Appendix A. Integrals

A.1. A logarithmic integral. We show $\int_0^\pi \ln\sin u\,\mathrm{d}u = -\pi\ln 2$. Let $I_s = \int_0^\pi \ln\sin u\,\mathrm{d}u$. By symmetry about $\pi/2$, $I_s = 2\int_0^{\pi/2} \ln\sin u\,\mathrm{d}u$, and by $u \to \pi/2 - u$ also $I_s = 2\int_0^{\pi/2} \ln\cos u\,\mathrm{d}u$. Adding,

$$2I_s = 2\int_0^{\pi/2} \ln\left(\sin u\cos u\right)\mathrm{d}u = 2\int_0^{\pi/2} \ln\frac{\sin 2u}{2}\,\mathrm{d}u = \int_0^{\pi} \ln\sin v\,\mathrm{d}v - \pi\ln 2 = I_s - \pi\ln 2, \tag{A.1}$$

with $v = 2u$. Hence $I_s = -\pi \ln 2$. Equivalently $\int_0^\pi \ln|\cos t|\,\mathrm{d}t = -\pi \ln 2$.

A.2. Entropy of the arcsine law. For $w(j) = 1/(\pi\sqrt{\gamma^2 - j^2})$ on $(-\gamma, \gamma)$ put $j = \gamma\cos\beta$; then $w\,\mathrm{d}j = \mathrm{d}\beta/\pi$, uniform on $(0, \pi)$, and $\gamma^2 - j^2 = \gamma^2 \sin^2\beta$. Hence

$$-\int w \ln w\,\mathrm{d}j = \ln\pi + \frac{1}{\pi}\int_0^\pi \ln(\gamma\sin\beta)\,\mathrm{d}\beta = \ln\pi + \ln\gamma - \ln 2 = \ln\frac{\pi\gamma}{2}, \tag{A.2}$$

using A.1.

A.3. The phase average. Split $\frac{1}{2\pi}\int_0^{2\pi}(1+\cos\Phi)\ln(1+\cos\Phi)\,\mathrm{d}\Phi = I_1 + I_2$. With $1 + \cos\Phi = 2\cos^2(\Phi/2)$ and A.1,

$$I_1 = \frac{1}{2\pi}\int_0^{2\pi} \ln(1+\cos\Phi)\,\mathrm{d}\Phi = \ln 2 + \frac{2}{\pi}\int_0^\pi \ln|\cos t|\,\mathrm{d}t = \ln 2 - 2\ln 2 = -\ln 2. \tag{A.3}$$

Integrating by parts with $\cos\Phi\,\mathrm{d}\Phi = \mathrm{d}\sin\Phi$ (the boundary term vanishes, including at $\Phi = \pi$ where $\sin\Phi\ln(1+\cos\Phi) \to 0$),

$$I_2 = \frac{1}{2\pi}\int_0^{2\pi} \cos\Phi\ln(1+\cos\Phi)\,\mathrm{d}\Phi = \frac{1}{2\pi}\int_0^{2\pi} \frac{\sin^2\Phi}{1+\cos\Phi}\,\mathrm{d}\Phi = \frac{1}{2\pi}\int_0^{2\pi}(1-\cos\Phi)\,\mathrm{d}\Phi = 1. \tag{A.4}$$

The sum is $1 - \ln 2$.

A.4. The elliptic integral near the separatrix. For $k \to 1$, $\mathsf{K}(k) = \ln(4/k') + O(k'^2 \ln k')$ with $k' = \sqrt{1-k^2}$, the standard logarithmic singularity [13]; it is the divergence of the classical period at the separatrix used in Sec. 7.

Appendix B. Numerical methods and verification

All numerics use $\hbar = I = 1$. Sector Hamiltonians (68) are diagonalised densely with truncation $|m_i| \le N$, convergence being checked by varying N and requiring the weight on boundary states to be below 10^{-6}. For the pendulum sectors we use $N = 8\sqrt{\lambda} + 150$ and diagonalise separately in the two parity sectors (Sec. 6). Where exponentially degenerate exchange doublets are returned as mixtures, the branch entropy is obtained from the weight on one side of $m_1 = M/2$, renormalised. Classical cell weights (90) are computed by adaptive quadrature, with the band-edge singularities supplied to the routine; for the arcsine law they are evaluated in closed form, $w_j = \pi^{-1}[\arcsin(j_+/\gamma) - \arcsin(j_-/\gamma)]$ with $j_\pm$ the cell edges clipped to $[-\gamma, \gamma]$. Bessel weights are summed over $|j| \le \gamma + 40\gamma^{1/3} + 50$, where the neglected tail is below 10^{-10}. Table B.1 collects the comparisons quoted in the text.
The scripts that produce every figure and table are available from the authors.

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Table B.1: Analytic results of this paper against exact numerics ($\hbar = I = 1$). Unless stated, perturbative checks are at $n = 7$ ($m_1 = 9$, $m_2 = 2$) and $\lambda = 0.02$; Mathieu checks at $q = 14.6$.

quantity	analytic	numerical
c_+, Eq. (35)	-0.0012500	-0.0012500
c_-, Eq. (35)	0.0016667	0.0016667
c_{++}, Eq. (40)	6.944444×10^{-7}	6.944456×10^{-7}
c_{--}, Eq. (40)	1.666667×10^{-6}	1.666669×10^{-6}
$E^{(2)}$, Eq. (37)	4.166667×10^{-6}	4.166668×10^{-6}
$p_0 = 1 - \varepsilon_+ - \varepsilon_-$ at $\lambda = 0.4$	0.9982639	0.9982649
S at $n = 0$, $\lambda = 0.02$, Eq. (57)	0.0020420	0.0020411
S at $n = 30$, $\lambda = 0.05$, Eq. (60)	2.108355×10^{-5}	2.114621×10^{-5}
$M = 0$ lowest level vs $a_0(q)/4$	-5.4542890	-5.4542890
$M = 0$ third level vs $a_2(q)/4$	1.3368673	1.3368673
$M = 1$ lowest level vs $b_1(q)/4$	-5.4542787	-5.4542787
$M = 1$ second level vs $a_1(q)/4$	-1.9036729	-1.9036729
B, $E_{\rm rel} = 300$, $\lambda = 100$, Eqs. (85),(86)	0.0927037	0.0927037
B, $E_{\rm rel} = 60$, $\lambda = 100$, Eqs. (85),(86)	0.1596085	0.1596085
$p_{j=0}$ at $n = 400$, $\gamma = 5$ vs $J_0^2(5)$	0.0315406	0.0314917
$p_{j=3}$ at $n = 400$, $\gamma = 5$ vs $J_3^2(5)$	0.1331018	0.1343519
arcsine entropy $\ln(\pi\gamma/2)$, $\gamma = 10^4$	9.6619231	9.6619231
$S_{\rm cl}^{\rm bin} - S_q$ at $\gamma = 10^6$ vs $1 - \ln 2$	0.3068528	0.2978385
S_0 at $\lambda I/\hbar^2 = 10240$, Eq. (118)	3.2067189	3.2067169
$N_{\rm lib}$ at $\lambda I/\hbar^2 = 800$ vs $(4/\pi)\sqrt{q}$	50.92958	51.00000

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